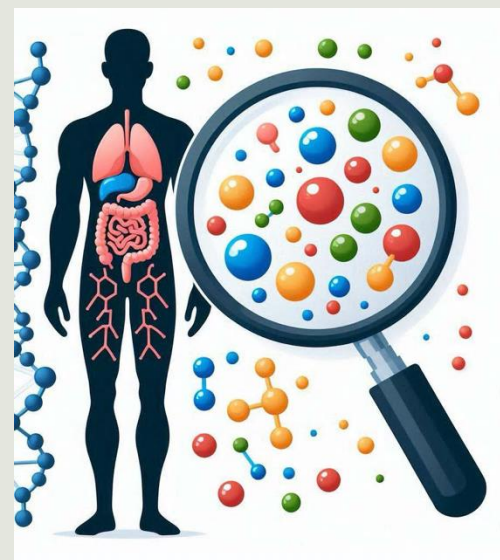
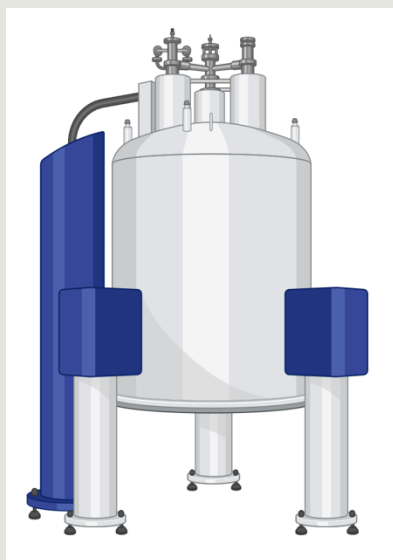


NMR in clinical metabolomics: an outdated tool?



Pascal de Tullio
Clinical Metabolomics group

Clinical Metabolomics: the analytical tools

LC(GC)-MS

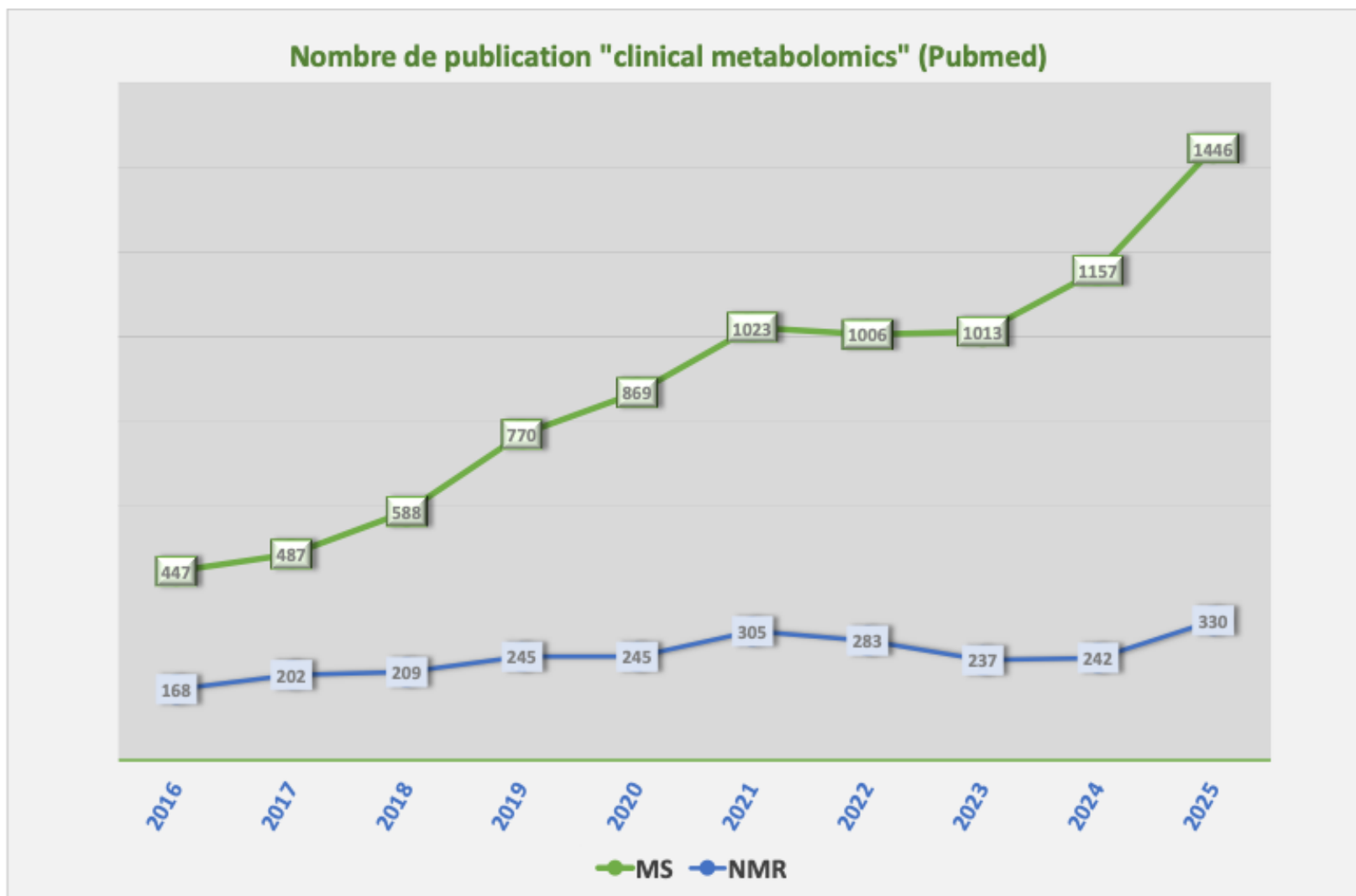


NMR



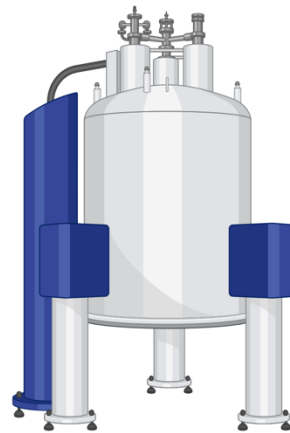
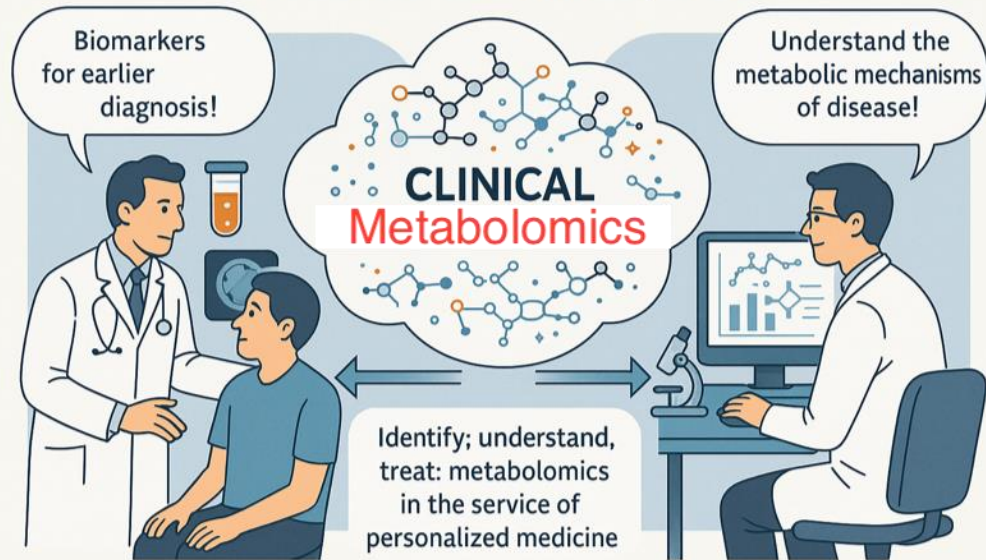
Others: Raman,
CE, ...

Clinical Metabolomics: the analytical tools

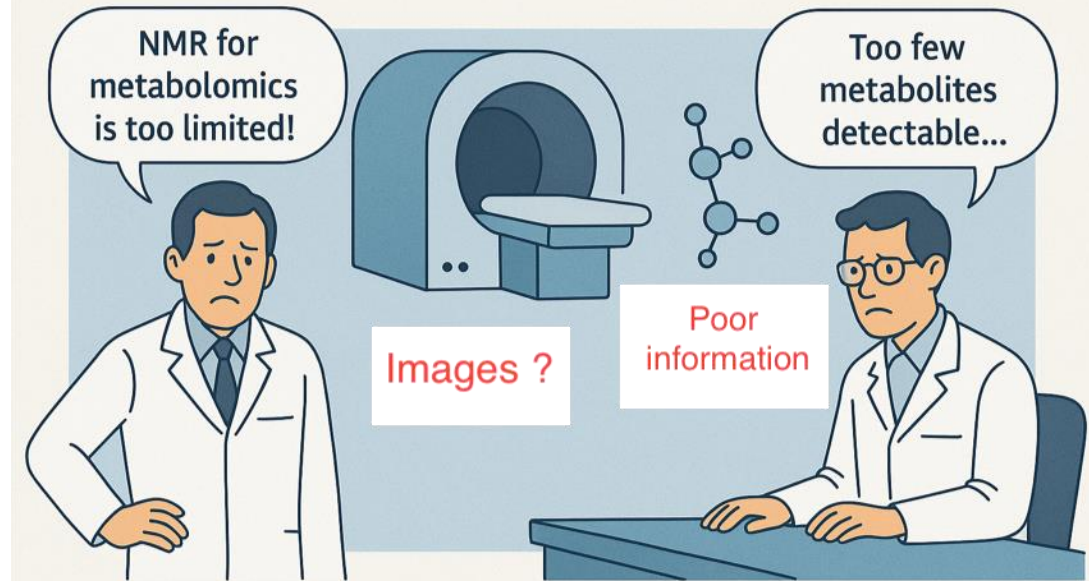


NMR in clinical Metabolomics

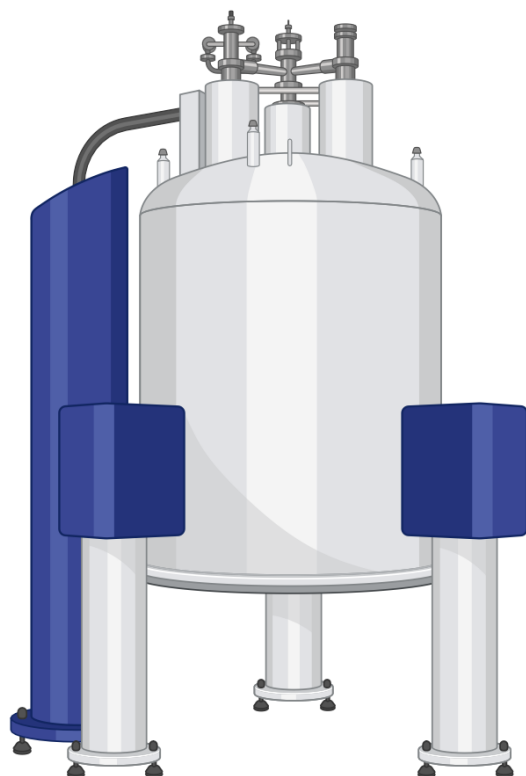
Metabolomics, at the intersection of research and clinical practice



Limitations of NMR in Metabolomics



NMR in clinical Metabolomics



Not an analytical
platform =
structural analysis

Poor sensitivity =
low number of
metabolites

Old and
complex
technology

Space-
consuming and
expensive

Poorly adapted to clinics

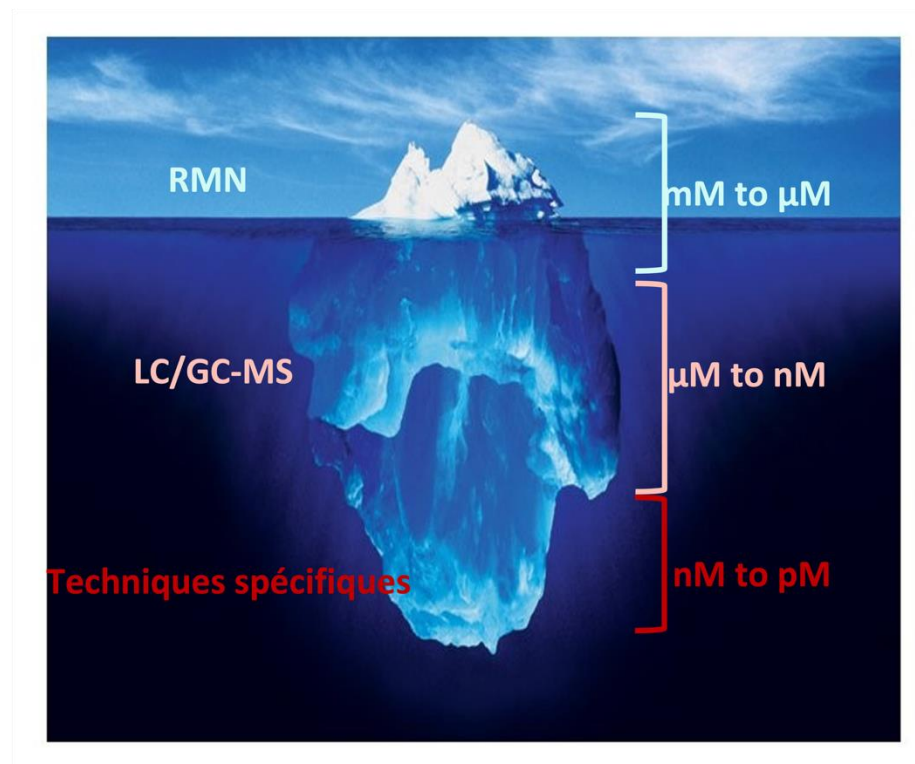
NMR : an interesting analytical platform with numerous qualities but ...

Reproducibility

Specificity

Quantification

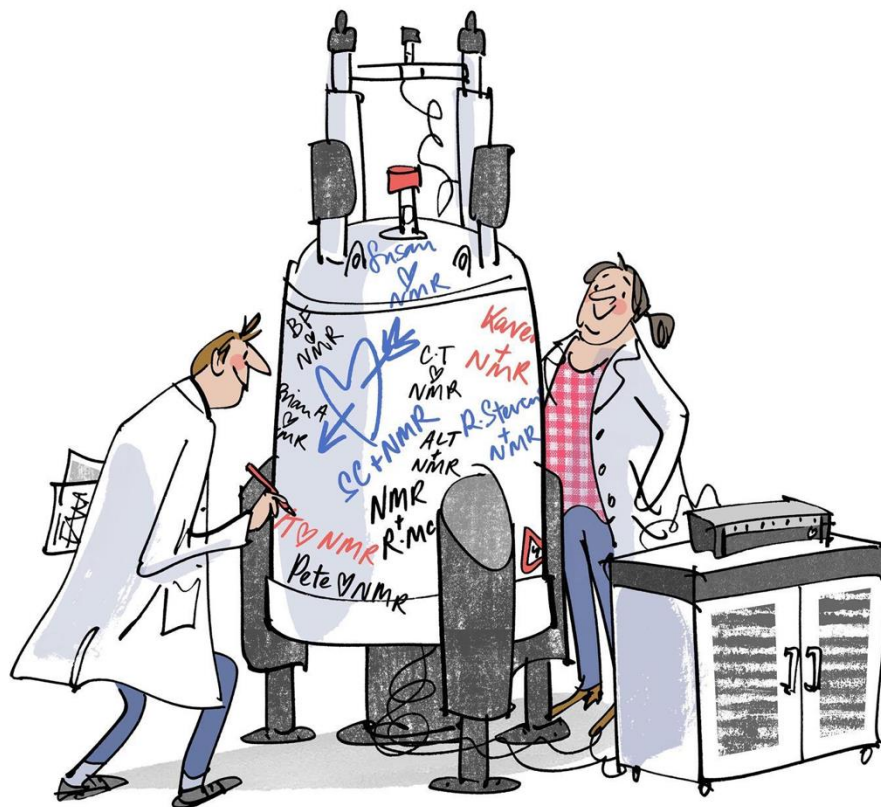
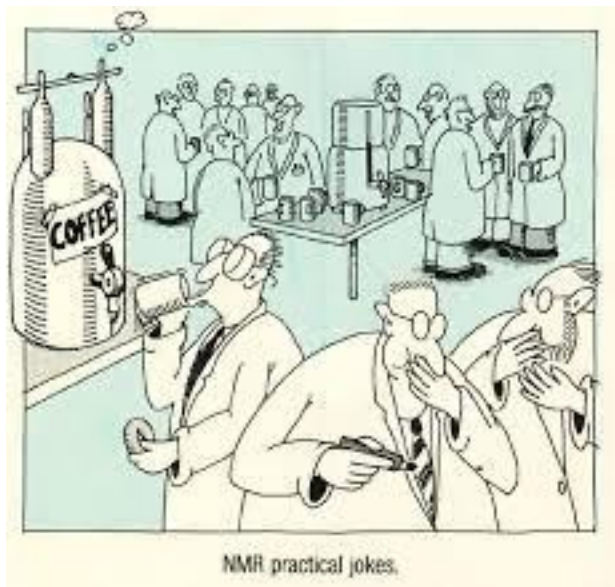
Automation



Sensitivity
Resolution

NMR in clinical Metabolomics

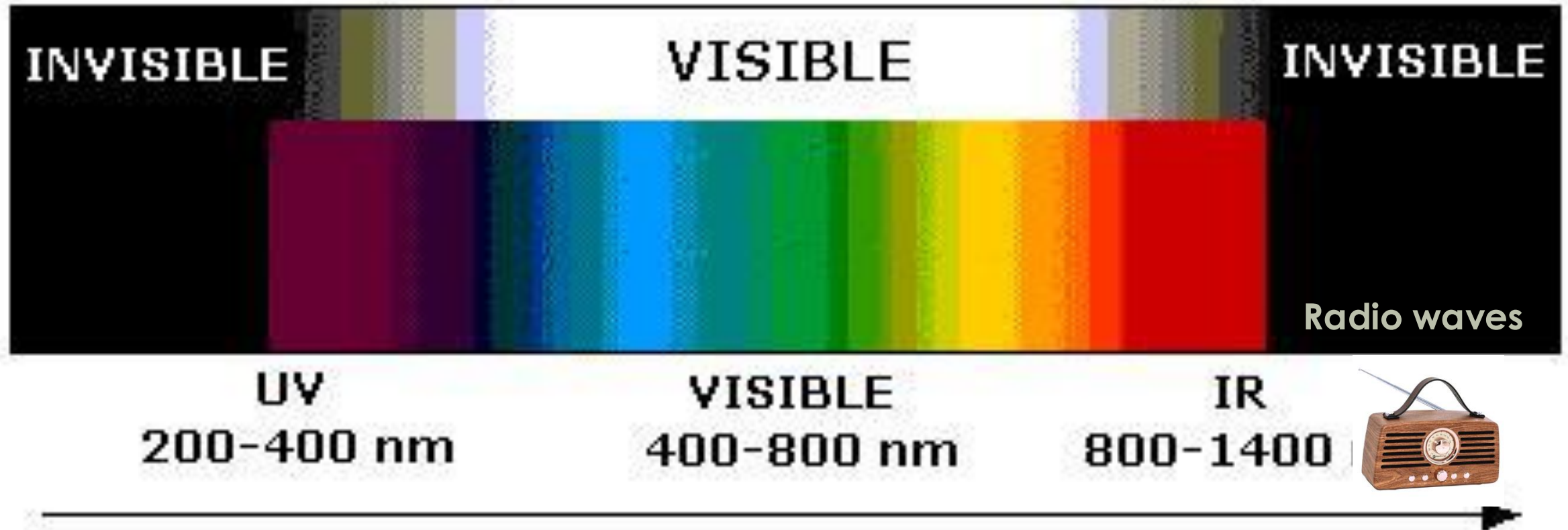
Is there a future for NMR in metabolomics ?



What is NMR ?



What is NMR ?



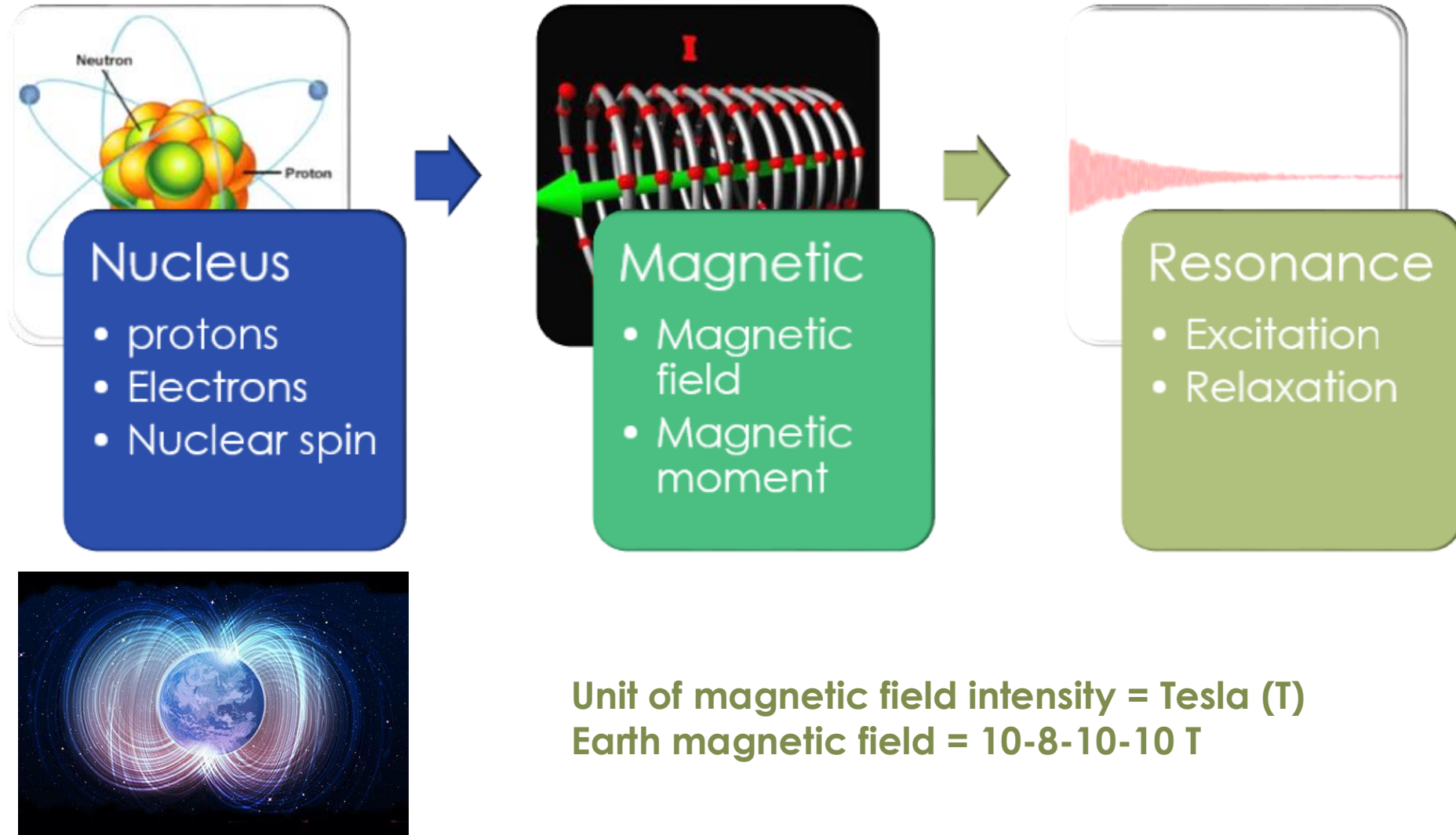
X-ray
diffraction

UV-visible
spectroscopy

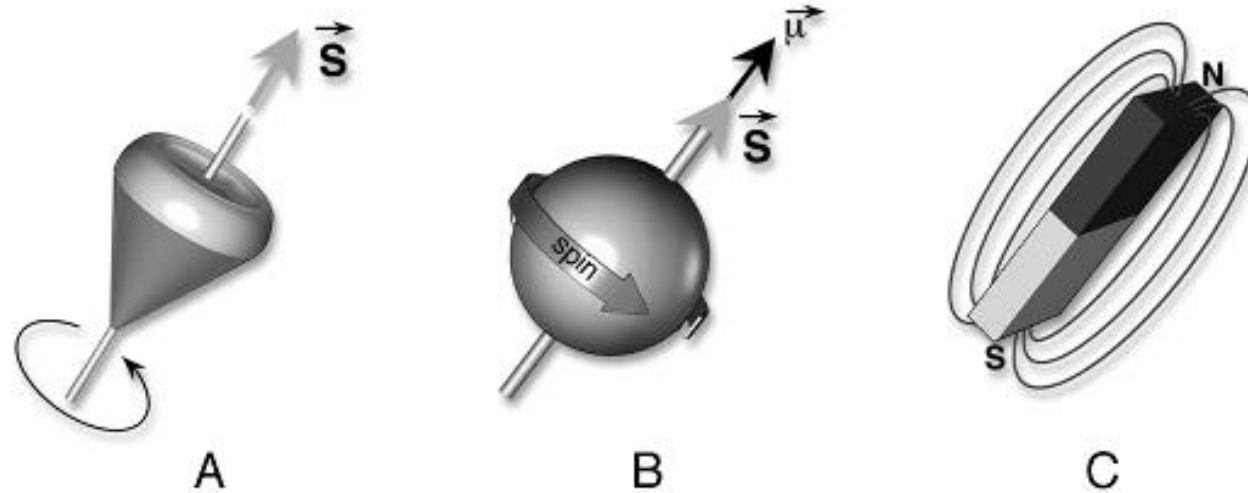
IR/Raman
spectroscopy

NMR
Spectroscopy

What is NMR ?

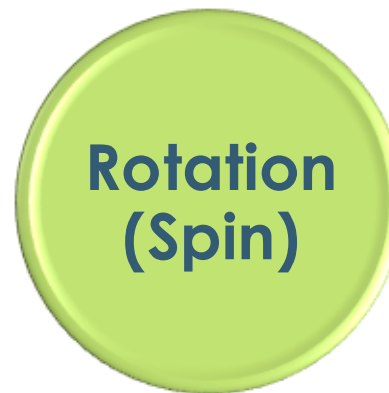


The N for Nucleus



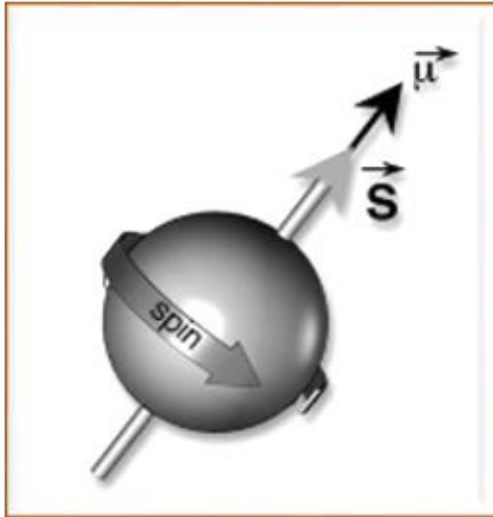
$\vec{S}$ = kinetic moment

$\vec{\mu}$ = magnetic moment

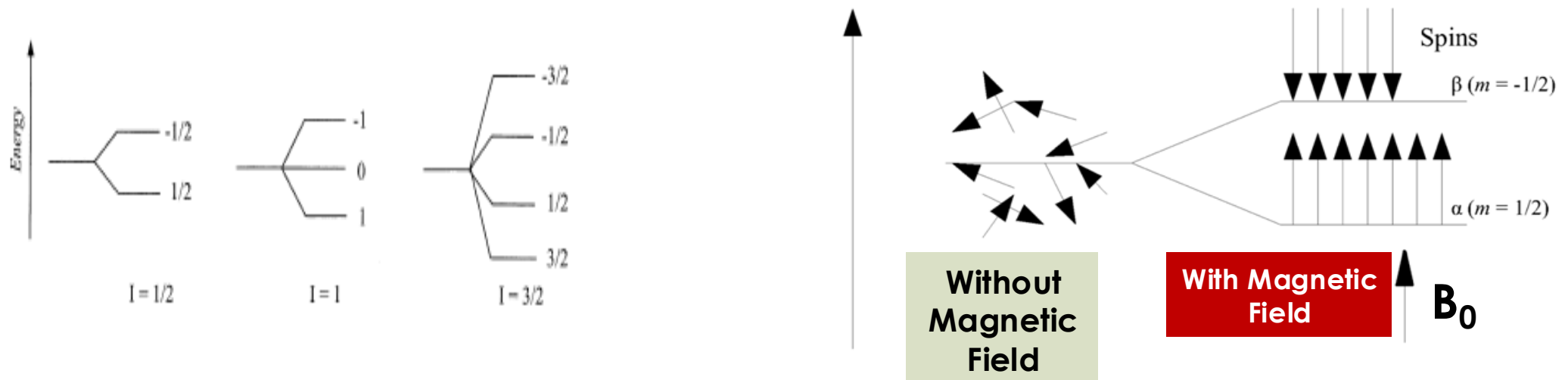


Magnetic
field
Magnetic
moment(μ)

The M for Magnetic

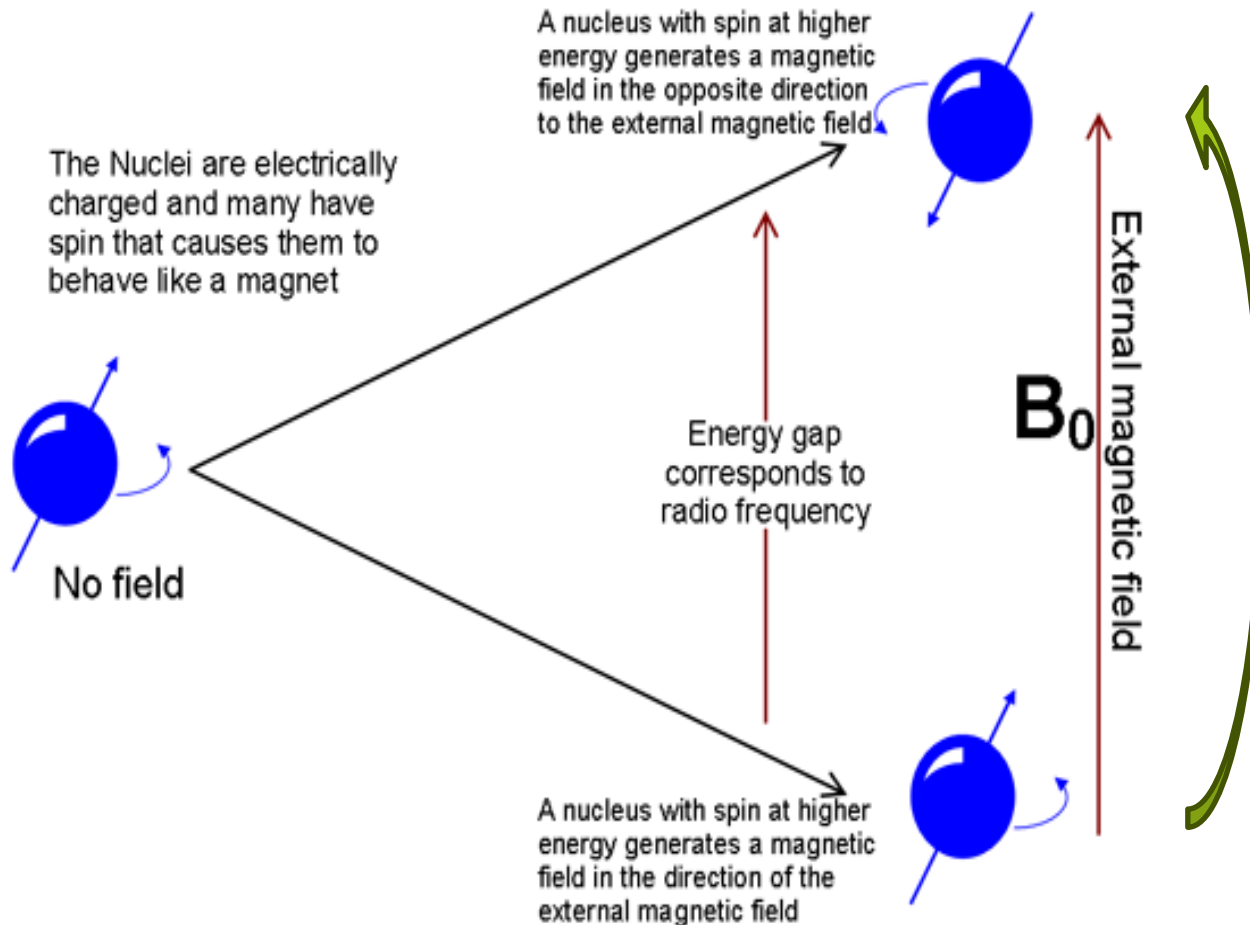


In quantum mechanics, μ is correlated with the spin number I ($\mu = \gamma I$) which defines the number of energy state that μ can take



The R for Resonance

The case of the spin- $\frac{1}{2}$ nucleus



NMR phenomena

=

Switch of spin from α to β

$$\Delta E = h\nu$$

$$\nu = B_0 \frac{\gamma}{2\pi}$$

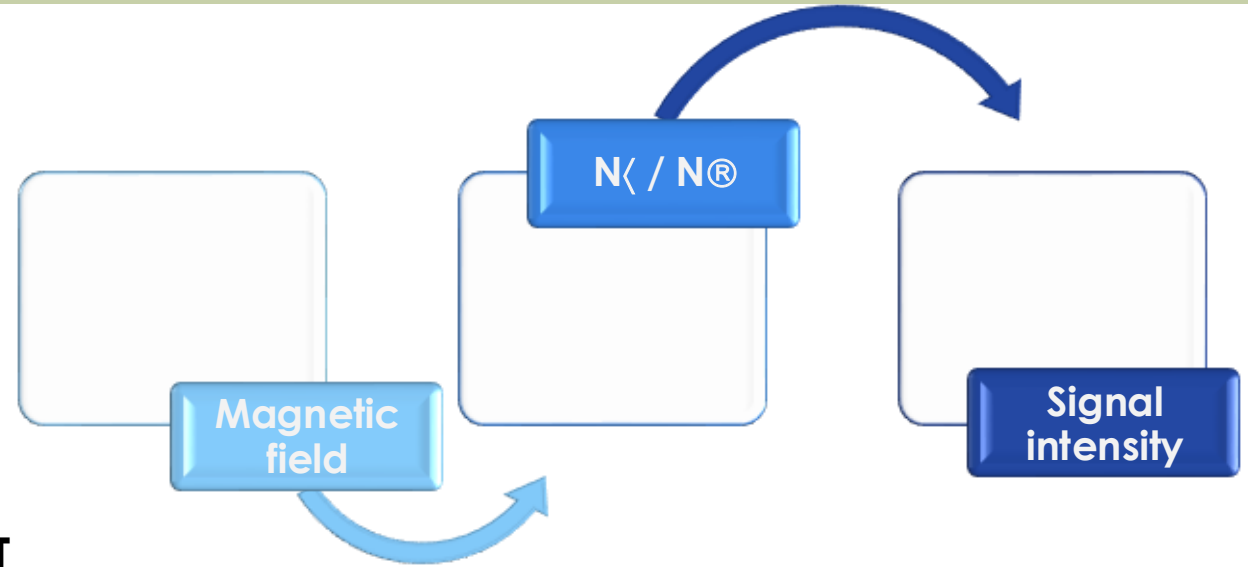
ν = Larmor Frequency

γ = Gyromagnetic constant

The R for Resonance

Boltzmann equation:

$$N_{\downarrow} / N_{\uparrow} = 1 + \frac{h B_0 \gamma}{2\pi kT}$$



Earth magnetic field: 10^{-8} T

$$N_{\downarrow} / N_{\uparrow} = 1 + \frac{h B_0 \gamma}{2\pi kT}$$



$$N_{\downarrow} / N_{\uparrow} \approx 1$$



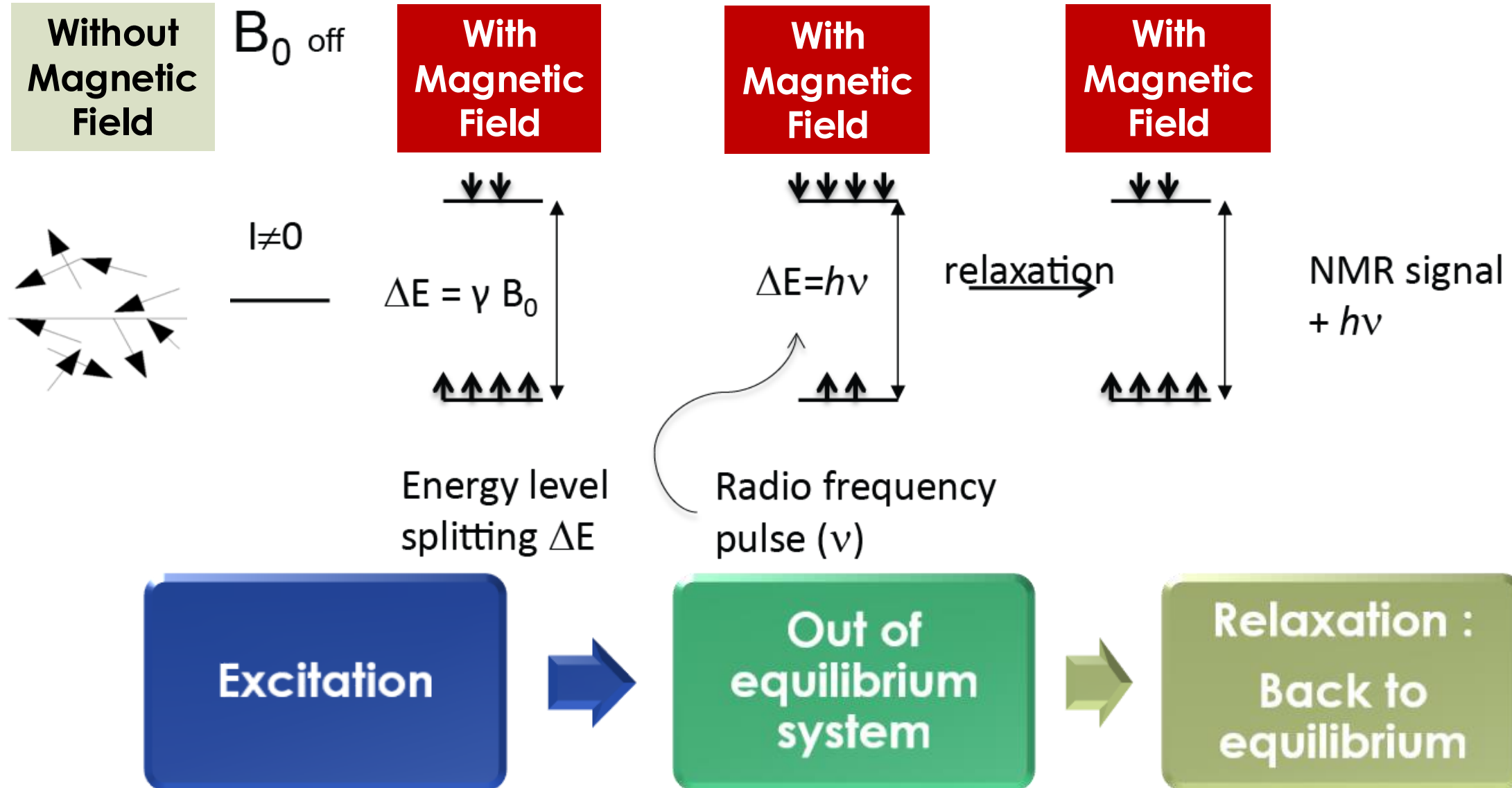
No NMR phenomena

7.050	9.40	11.75	14.09	16.44	17.62	Valeur du champ B_0 (Tesla)
300.130	400.130	500.130	600.130	700.130	750.130	Fréquence de Larmor (MHz)

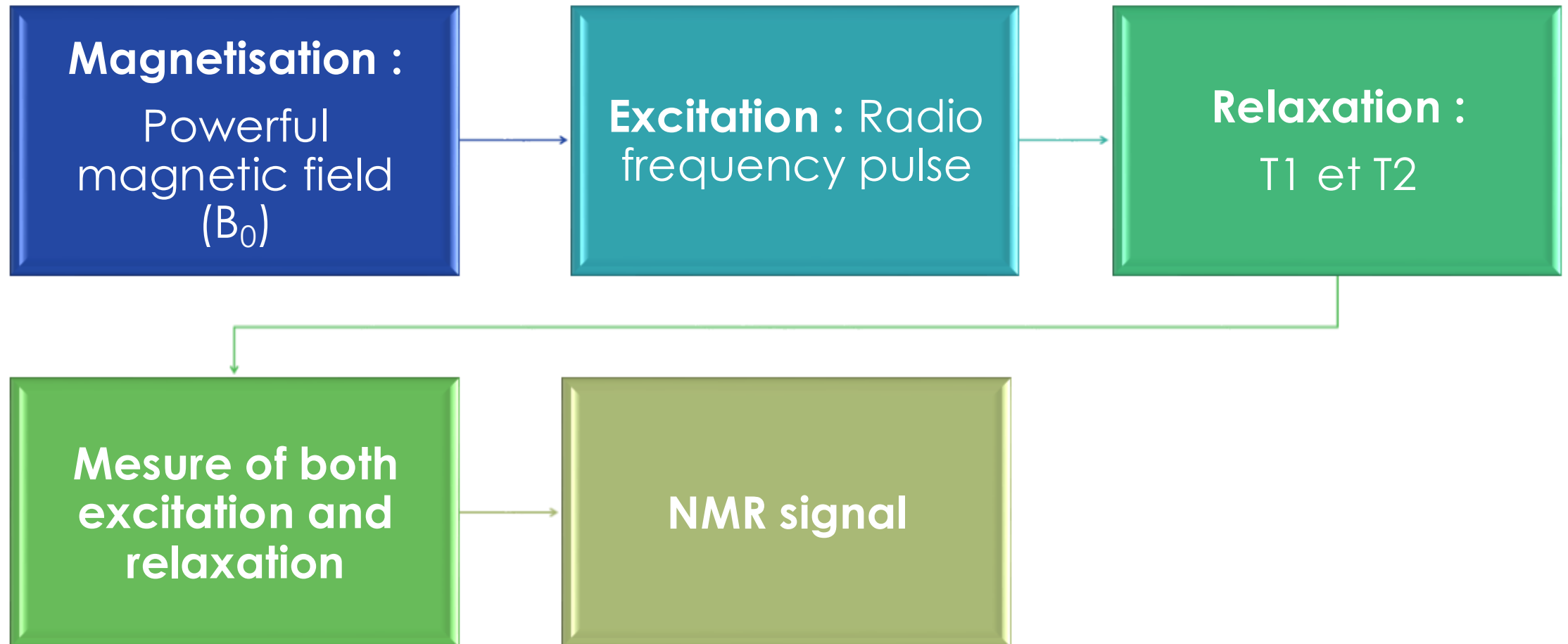


Powerful B_0

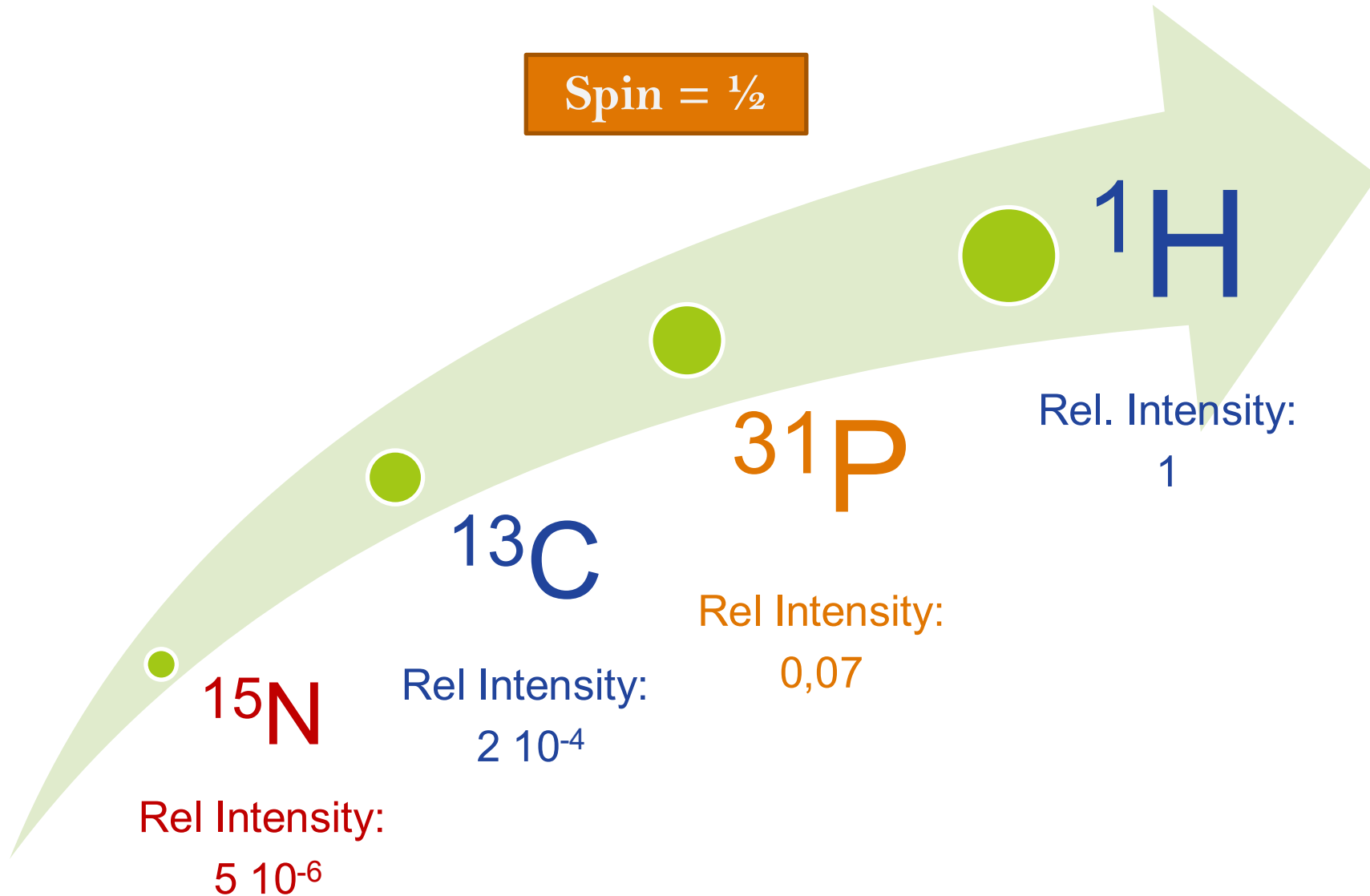
The M for Magnetic and the R for Resonance and Relaxation



The NMR process

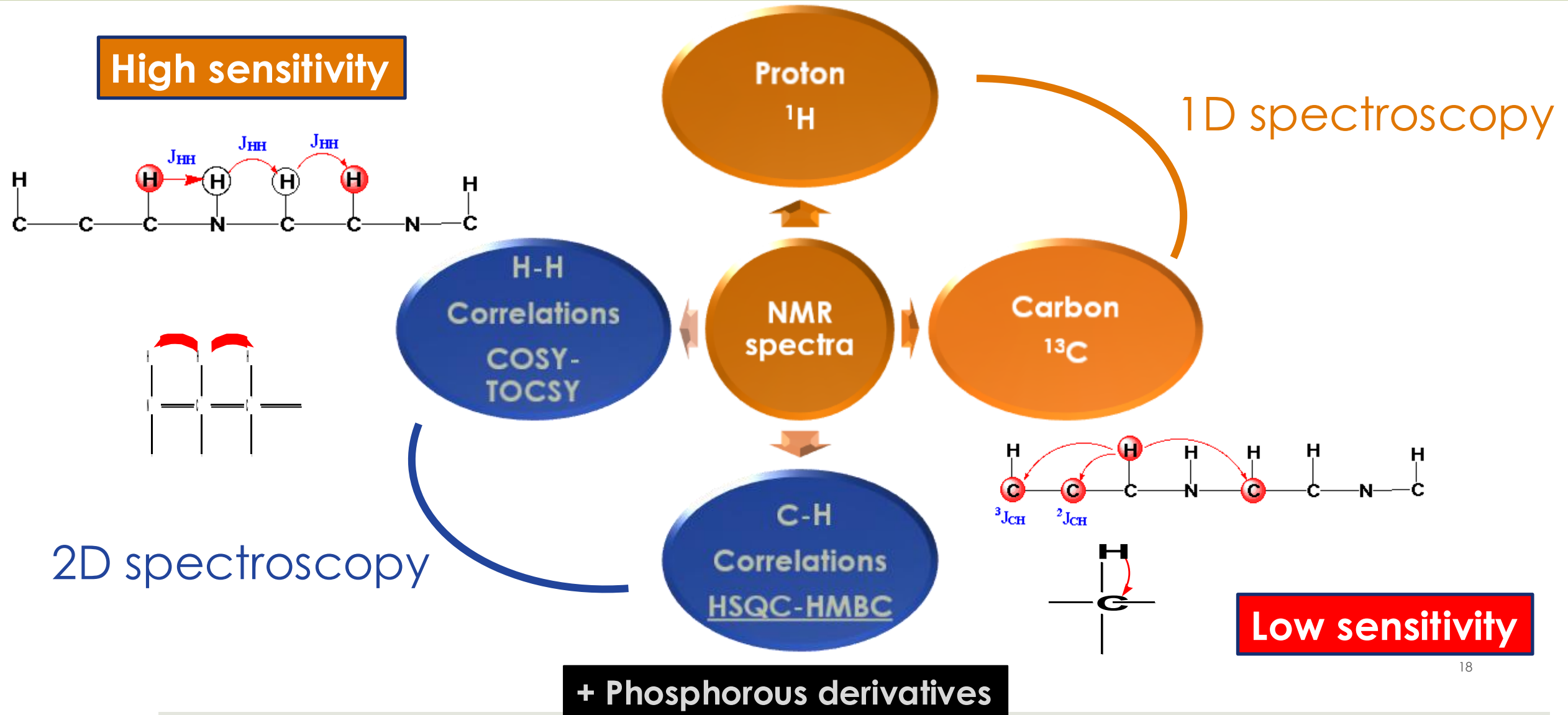


What is observable in NMR ?

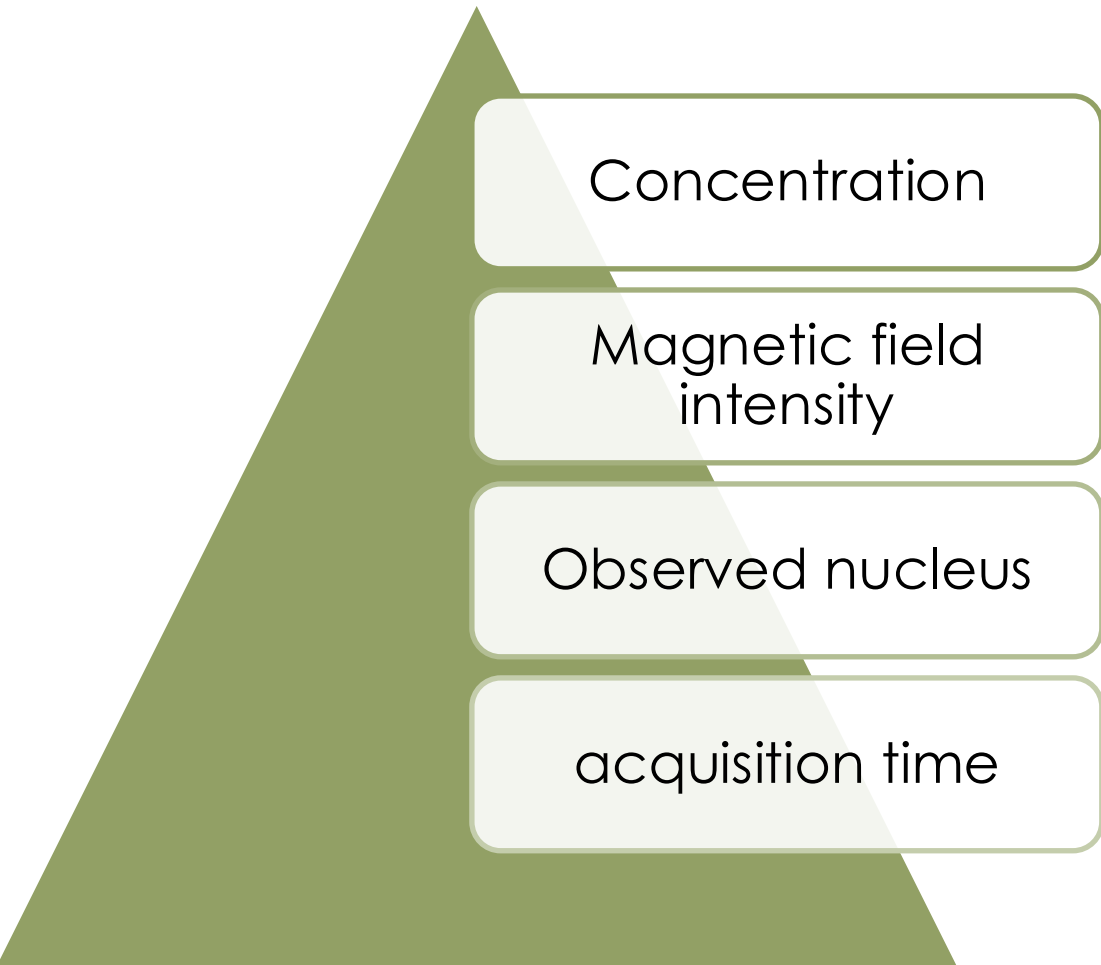


What is observable in NMR ?

High sensitivity



Intensities of signals in NMR



Concentration

Proportionality between concentration and signal intensities

Magnetic field intensity

Sensitivity increases with the magnetic field
700 MHz > 600 MHz > 500 MHz

Observed nucleus

$^1\text{H} > ^{31}\text{P} \gg ^{13}\text{C} \gg \gg ^{15}\text{N}$

acquisition time

Increasing acquisition time improve S/N

^1H NMR spectroscopy

Most adapted and used in metabolomics

Sensitivity is in the μM range

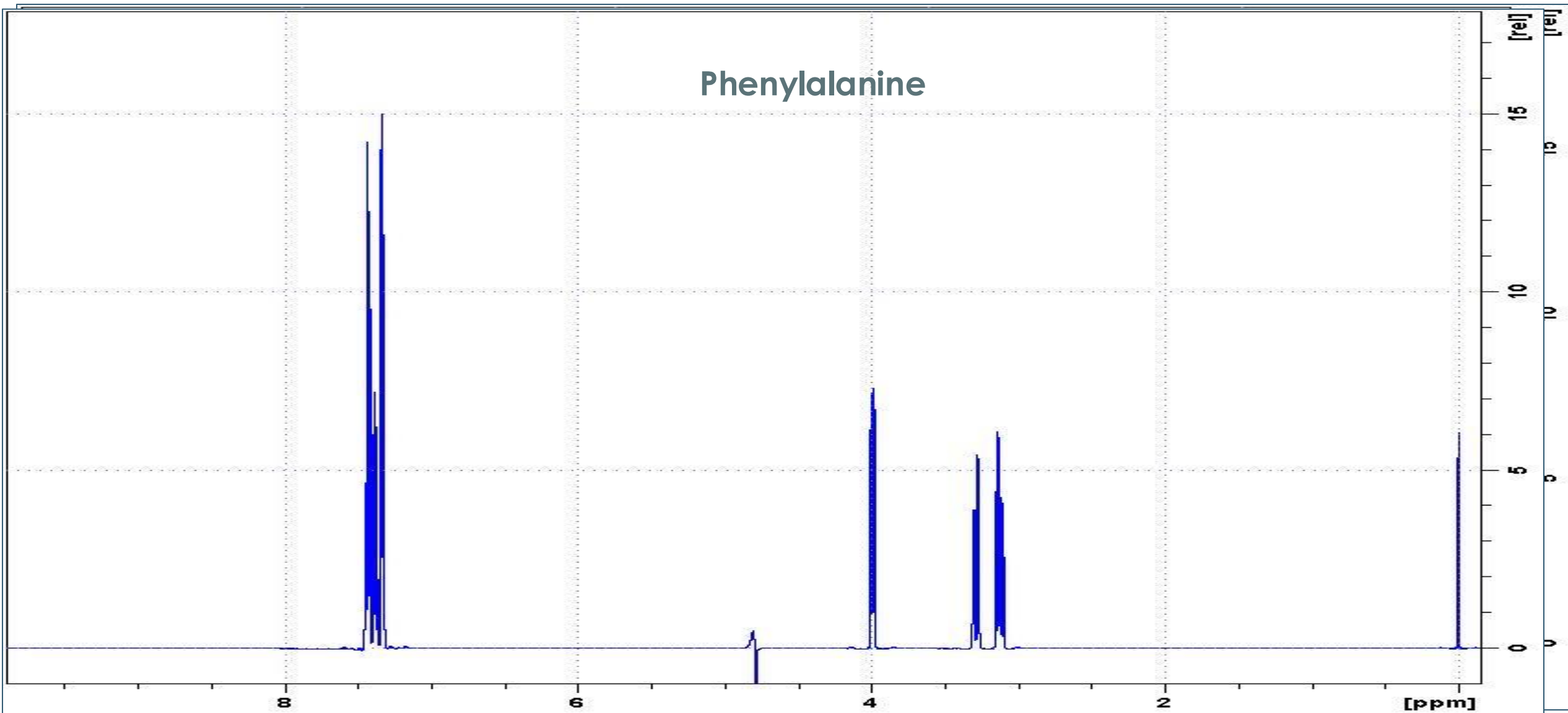
Quantitative

Poor time consuming and adapted to high-throughput: few minutes/samples

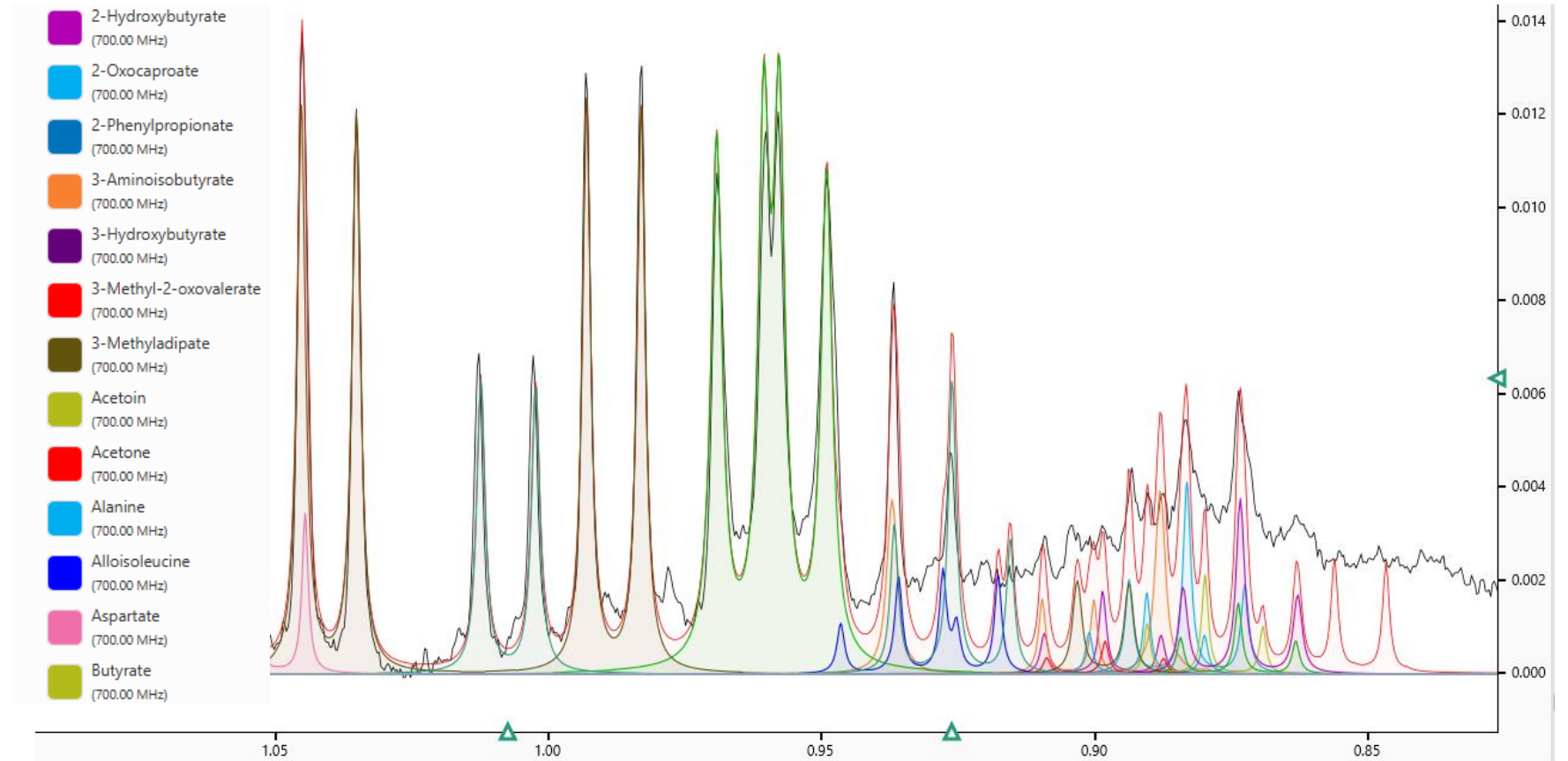
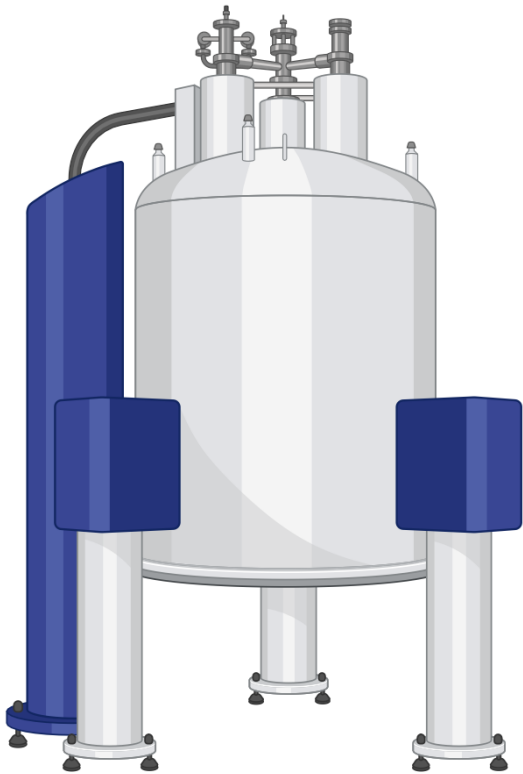
Each compound has a specific spectral signature with multiple signals

^1H NMR spectroscopy

Phenylalanine



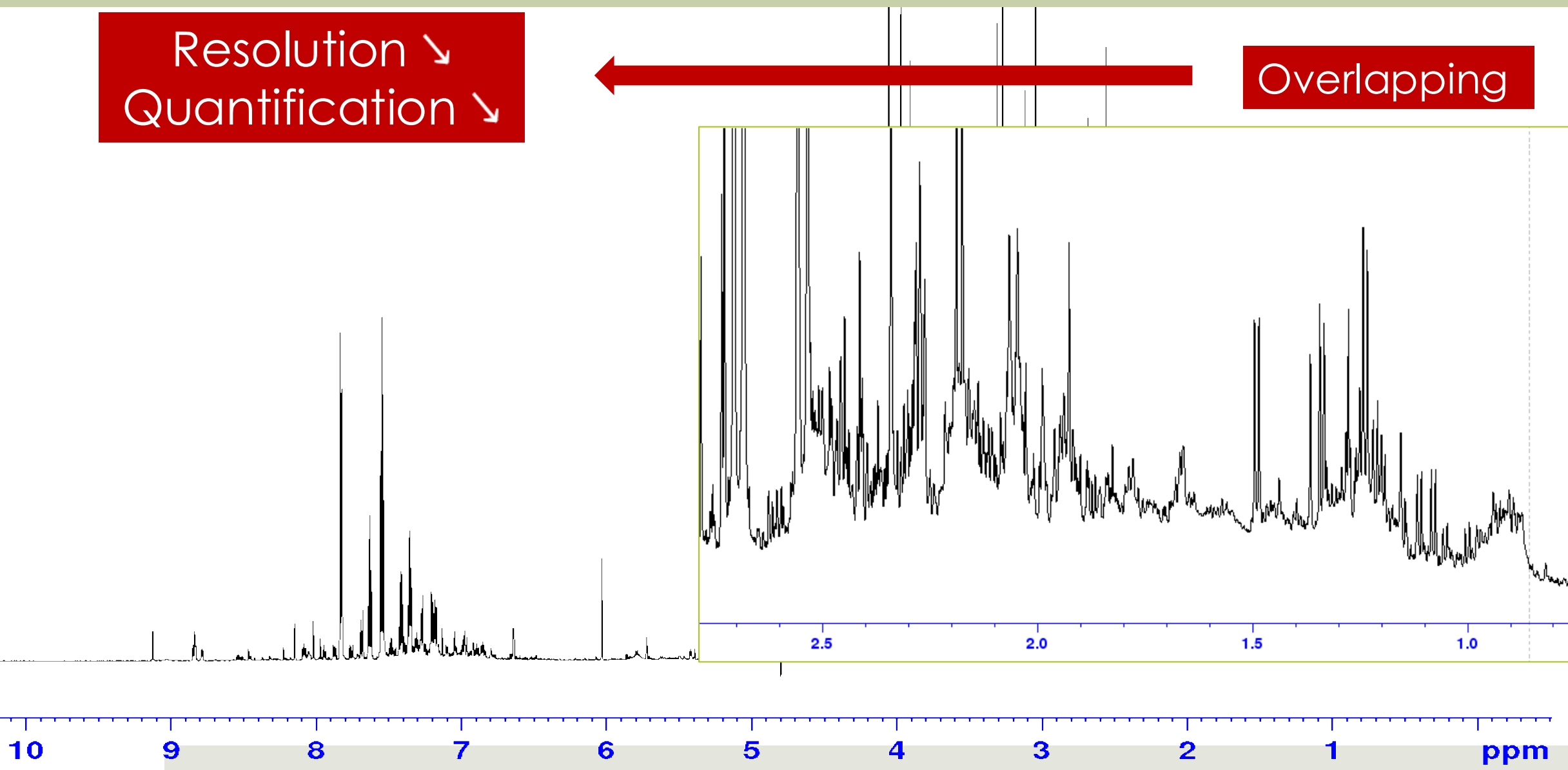
NMR of complex biological media



^1H NMR spectroscopy

Resolution ↘
Quantification ↘

Overlapping

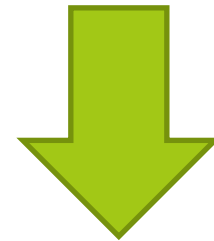


^1H NMR spectroscopy: the drawback

Sensitivity

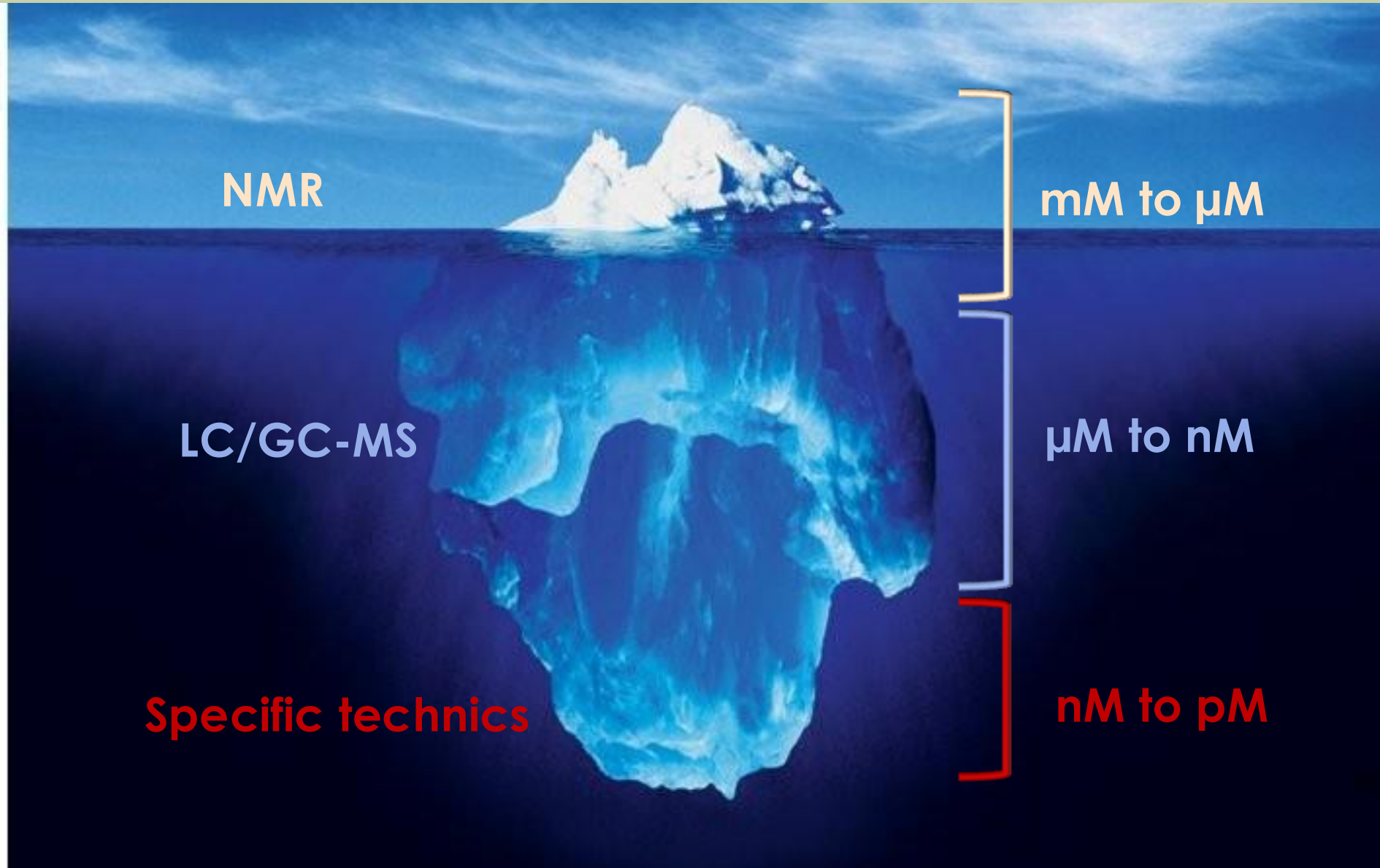


Resolution



1D ^{13}C
2D NMR

Metabolomics: the analytical tools



Metabolomics: the analytical tools

Generic
sample
préparation



Focused
sample
préparation

NMR

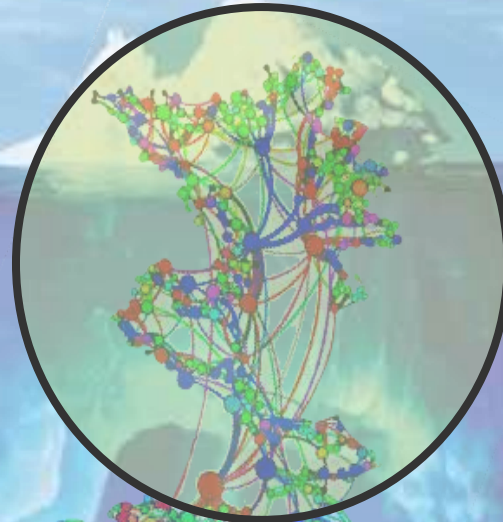
LC/GC-MS

Specific techniques

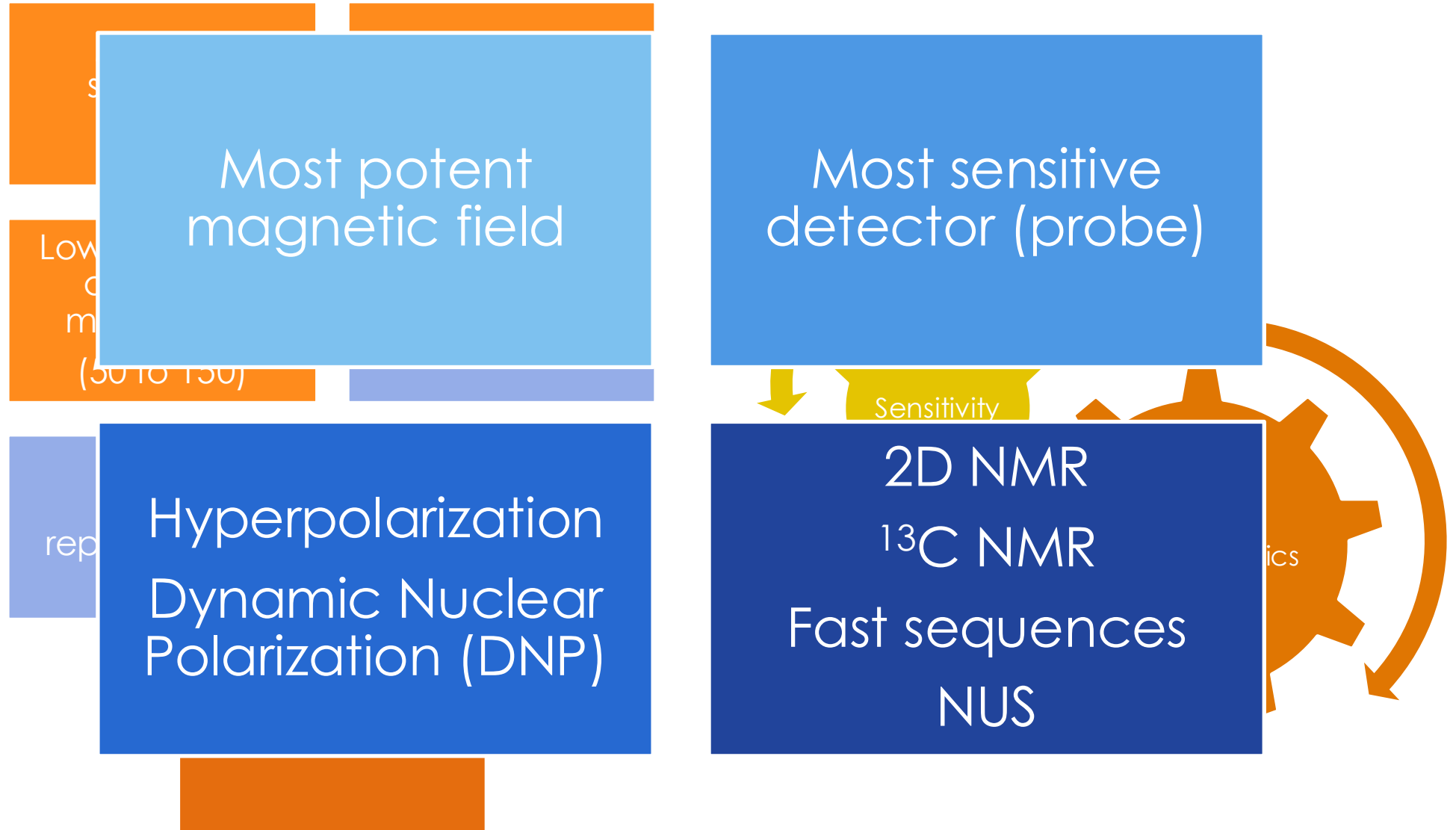
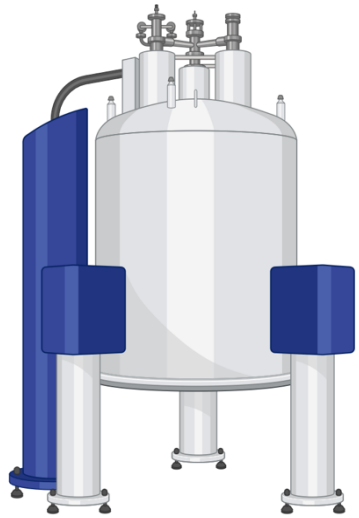
mM to μ M

μ M to nM

nM to pM



NMR in Metabolomics: Where Do We Stand Today?



Probe and Magnetic field



Increase the magnetic field

Current potency in routine: 500-600-700 MHz

Ultra-high field > 1 GHz

Increasing the field drastically increase the cost

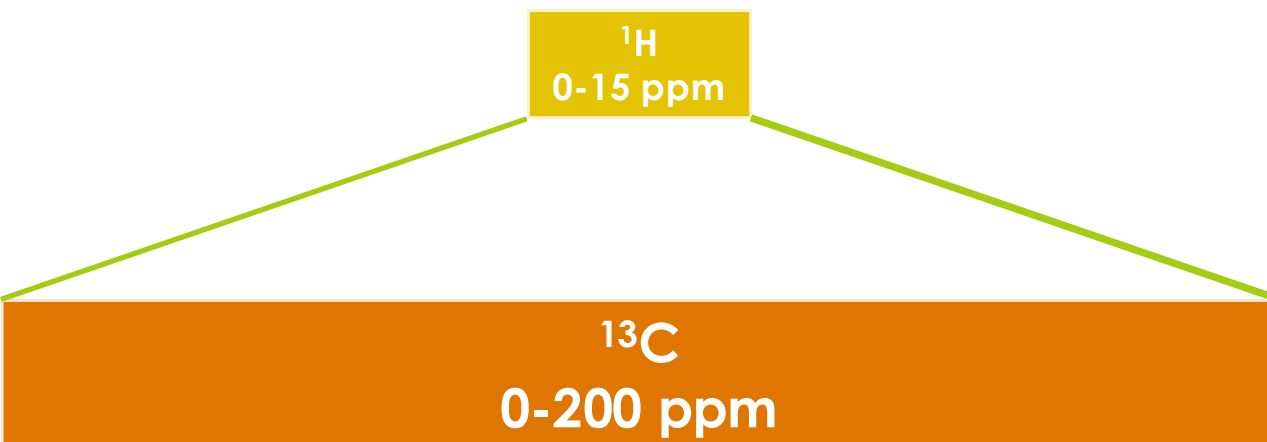


Improve the probe:
Cryoprobe

Gain > 4 for the S/N ratio

2D NMR and ^{13}C spectroscopy

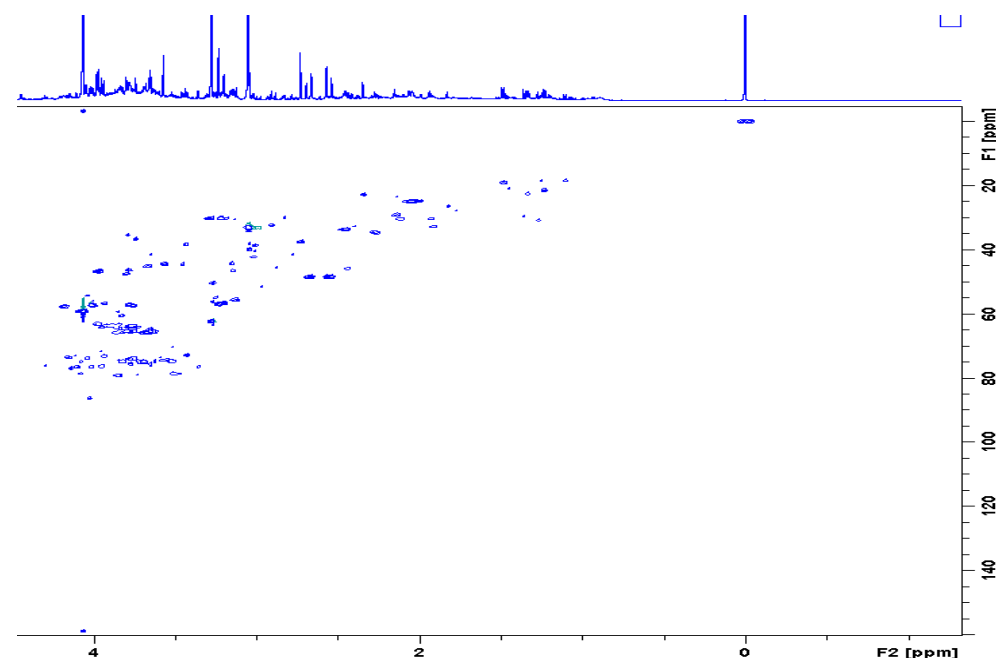
^{13}C spectroscopy



Huge increase in resolution
Huge decrease in sensitivity
No quantification

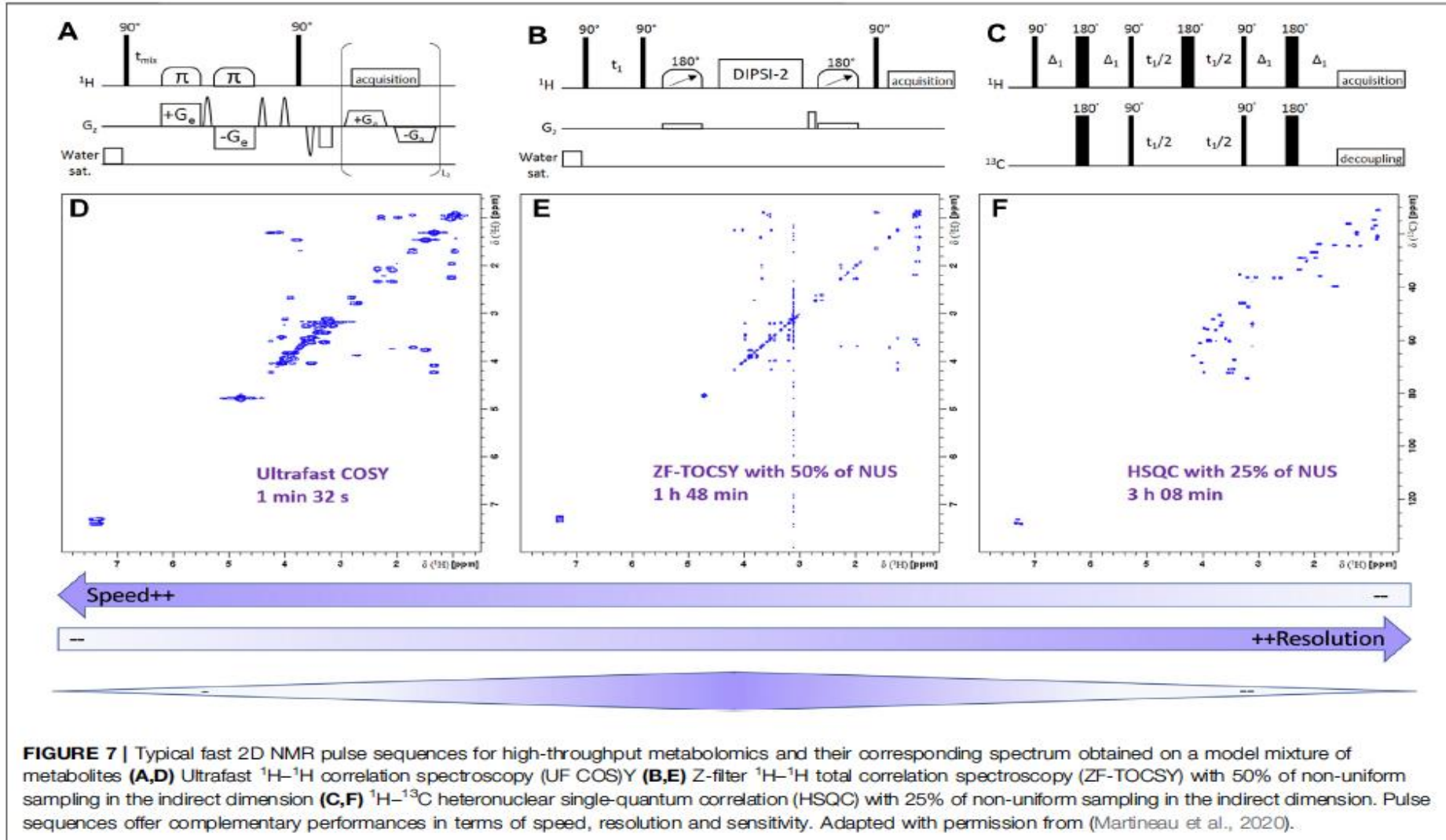
2D NMR spectroscopy

COSY-TCOSY-HSQC-HMBC



Huge increase in resolution
Decrease in sensitivity
Possible quantification
Timing consuming

2D NMR

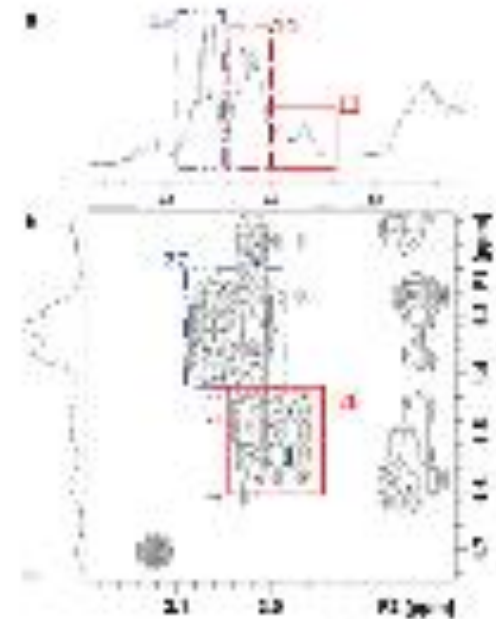
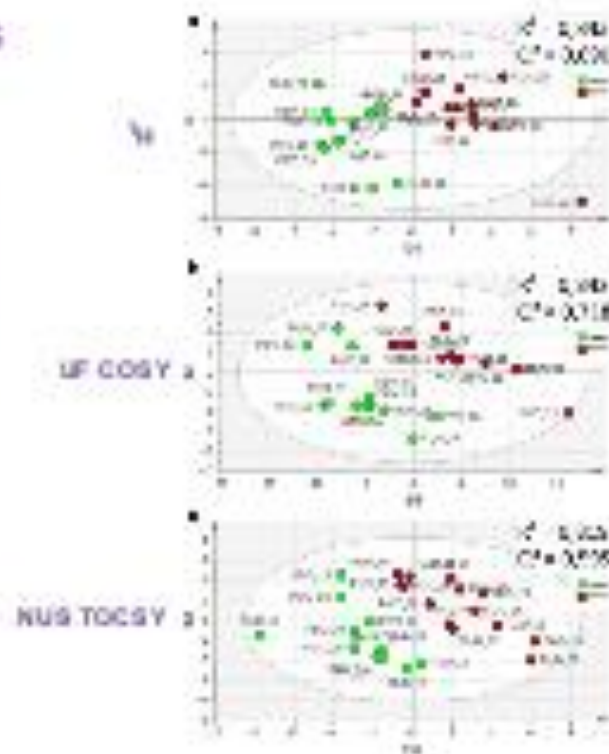
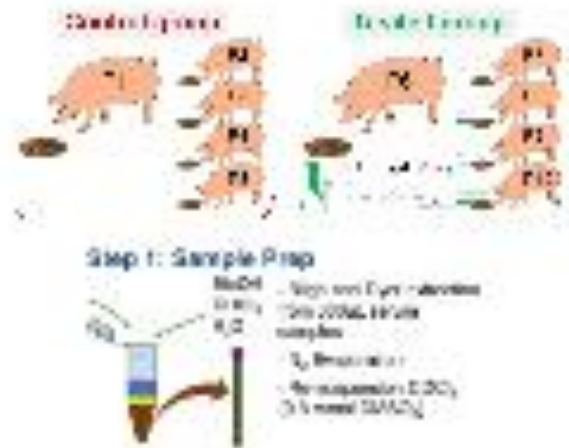


2D NMR

Where does NMR stand in metabolomics?

... keep overcoming the drawbacks of each analytical method, ...

2D NMR Methods

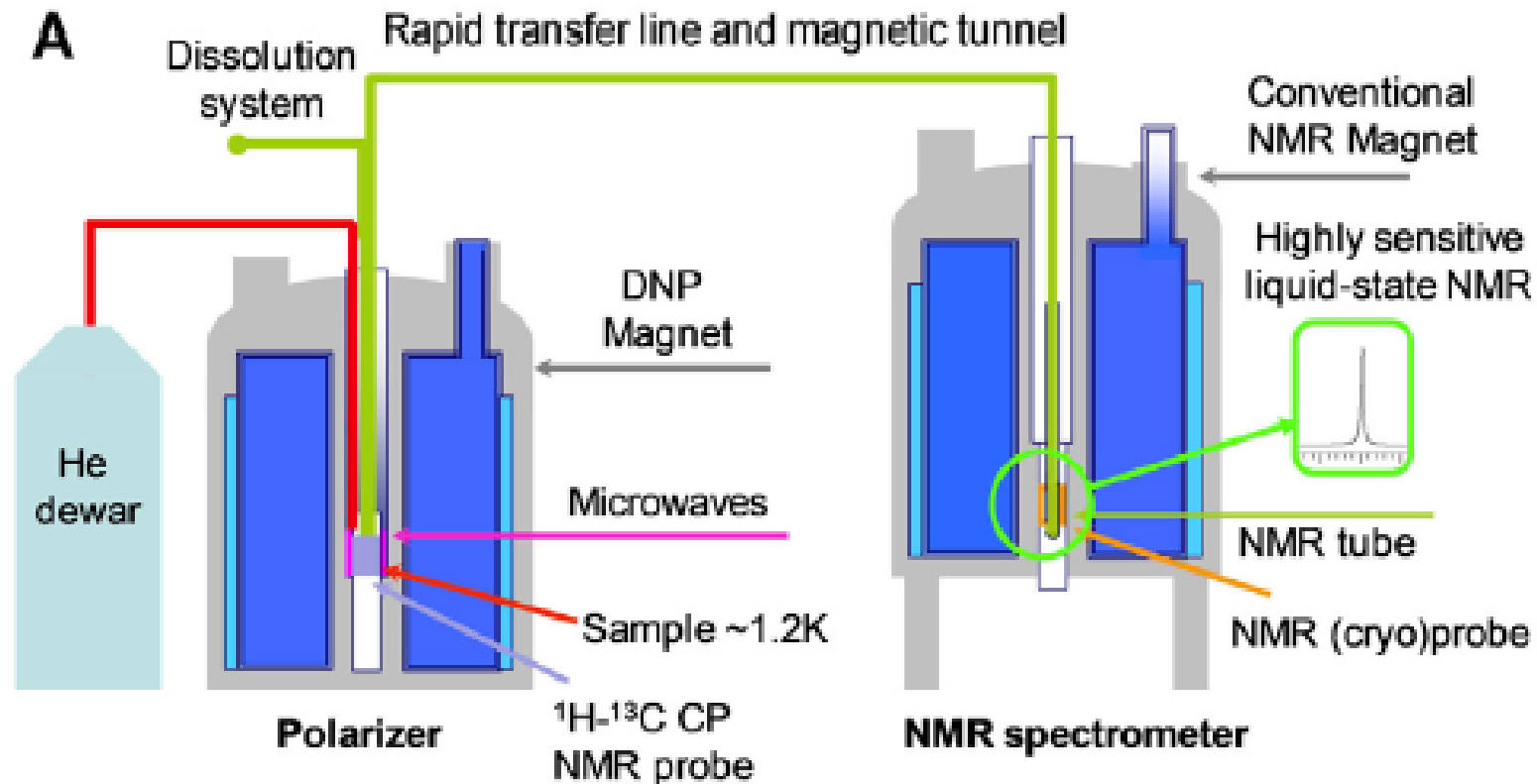


Marchant et al. Metabolomics, 2018

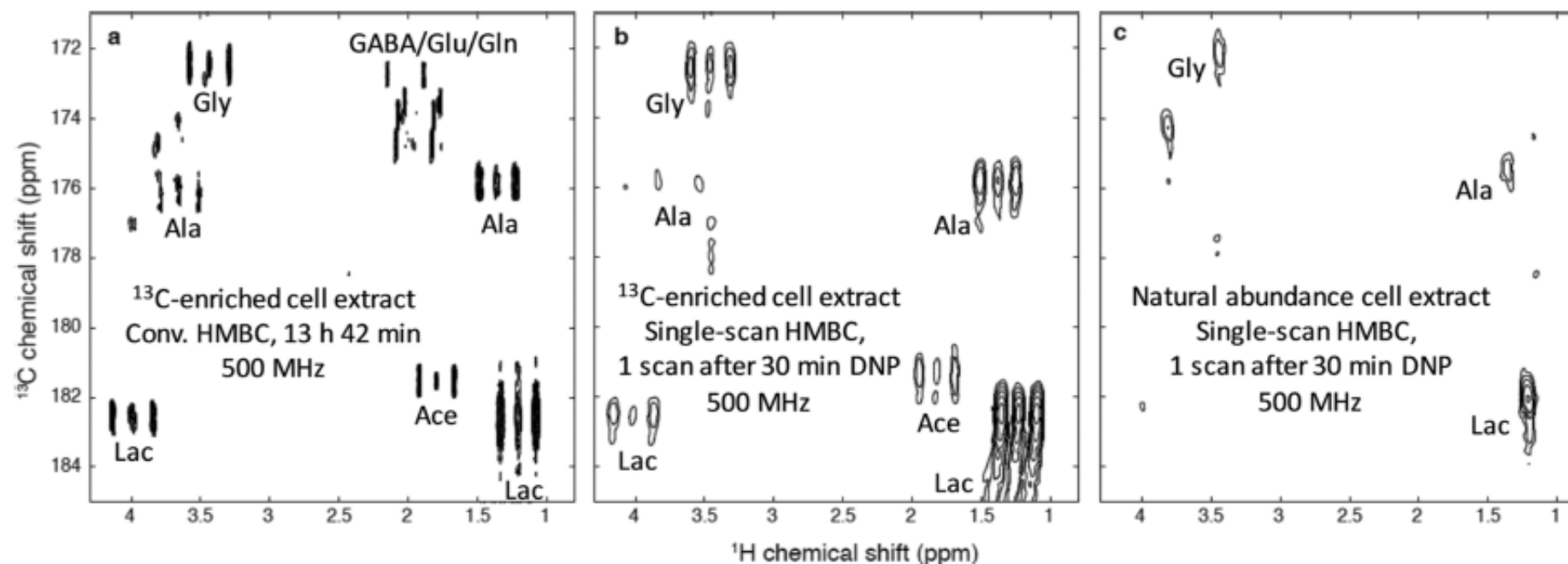
+ Improve the signal overlapping observed in conventional ^1H NMR metabolomics

Liquid DNP

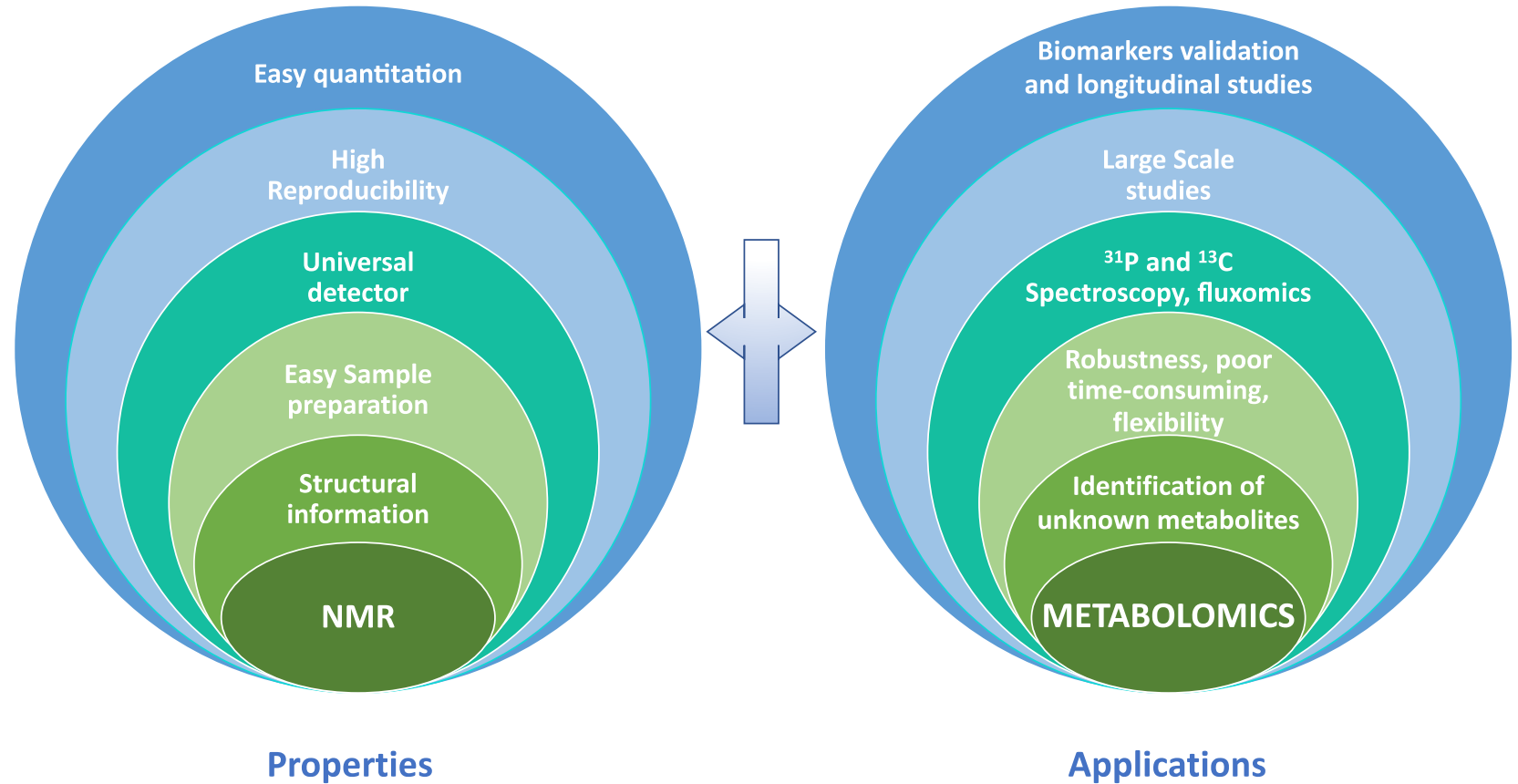
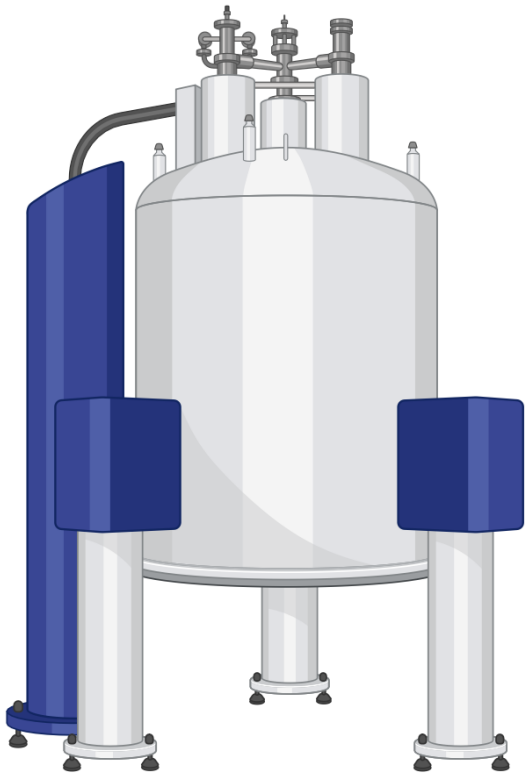
Improved sensitivity, especially for ^{13}C spectroscopy and 2D experiment
Still in development but very promising



Liquid DNP

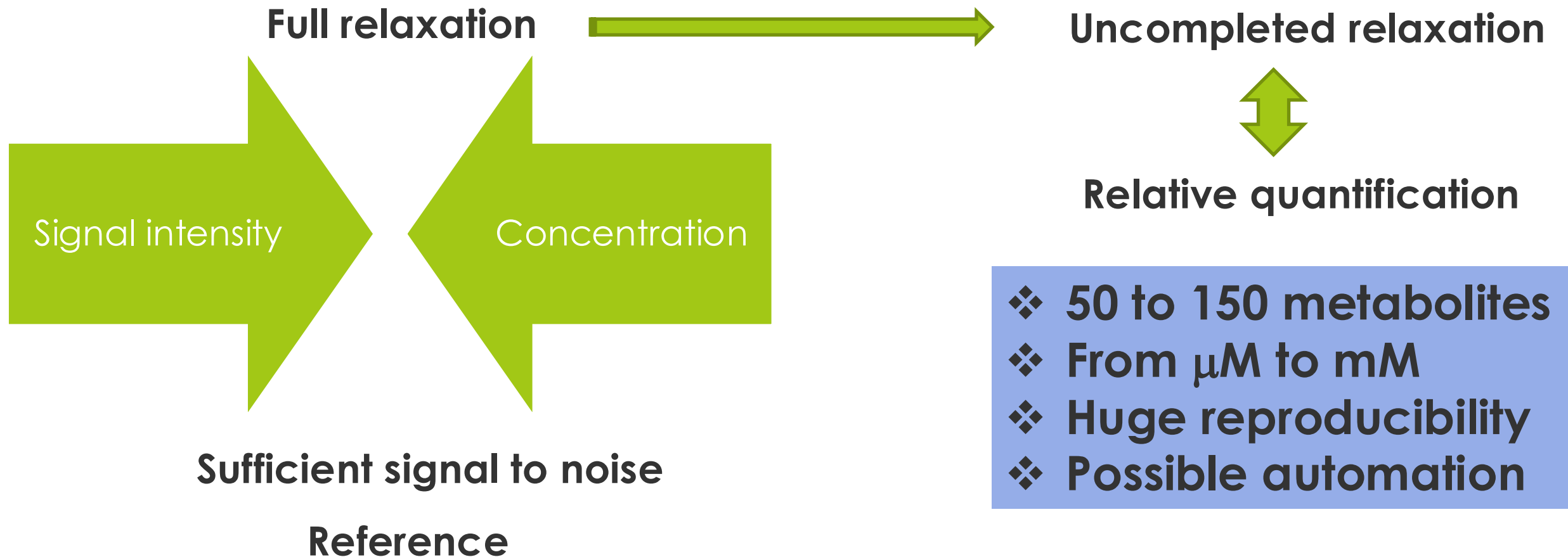


NMR in Metabolomics: Where Do We Stand Today?



NMR in Clinical Metabolomics : quantification

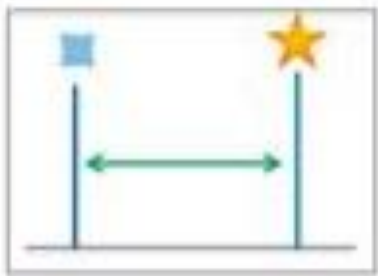
No biomarkers without quantification



NMR in Clinical Metabolomics : quantification

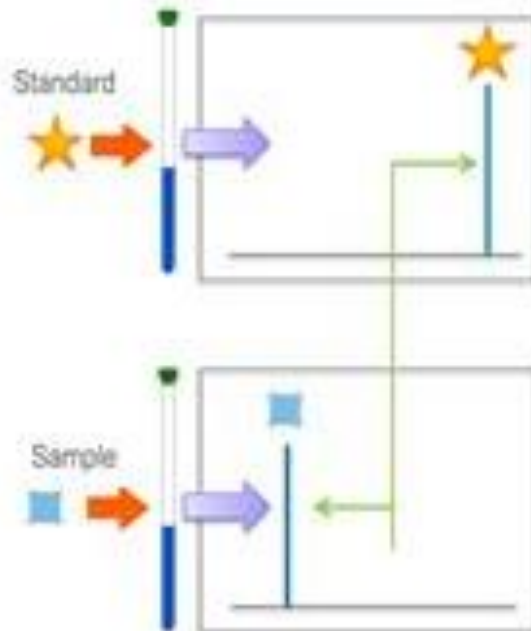
Absolute Quantitation

Internal Calibration (IC)



External Calibration (EC)

An example: pulse length-based concentration determination (PULCON)



Eq. (1)

$$\frac{I_A}{I_B} = \frac{H_A C_A}{H_B C_B}$$

Eq. (2)

$$P_{\text{sample}} = \frac{I_{\text{sample}} / H_{\text{sample}}}{I_{\text{std}} / H_{\text{std}}} \times \frac{M_{\text{sample}} / W_{\text{sample}}}{M_{\text{std}} / W_{\text{std}}} \times P_{\text{std}}$$

I = Signal intensity (Integral), H = Number of protons in a functional group, C = Concentration, W = Weight, M = Molecular weight, P = Purity (in %)

No calibration curve

No specific reference

Non-labelled reference

NMR in Clinical Metabolomics : quantification

Standards for ¹H quantitative NMR, TraceCERT®

PN	Substance	D ₂ O			CDCl ₃			DMSO-d ₆			CD ₃ OD			CD ₃ CN		
		δ (ppm)	T1 (s)	s (mg/ml)	δ (ppm)	T1 (s)	s (mg/ml)	δ (ppm)	T1 (s)	s (mg/ml)	δ (ppm)	T1 (s)	s (mg/ml)	δ (ppm)	T1 (s)	s (mg/ml)
01380	Ethylene carbonate	4.5	5.5	>250	4.5	7.0	>250	4.5	2.7	>250	4.5	5.3	>250	4.5	2.0	>250
03826	Calcium formate	8.4	3.4	>250	-	-	-	-	-	-	-	-	-	-	-	-
06185	Benzoic acid	-	-	-	8.1	3.7	150	8.0	3.3	>250	8.0	4.3	>250	8.0	4.5	80
					7.7	4.0		7.6	3.7		7.6	4.4		7.7	2.5	
					7.5	3.4		7.5	3.0		7.5	3.9		7.5	2.6	
06856	Duroquinone	-	-	-	2.0	3.3	>250	2.0	3.1	30	2.0	4.0	10	2.0	5.4	20
07038	Dimethyl terephthalate	-	-	-	8.1	3.6	160	8.1	2.9	20	8.1	4.4	4.0	8.1	4.9	20
					4.0	1.8		3.9	1.1		4.0	2.4		3.9	2.6	
14659	Potassium hydrogen phthalate	7.5	2.5	>250*	-	-	-	-	-	-	7.9	2.5	5.0	-	-	-
40384	1,2,4,5-Tetrachloro-3-nitro-benzene	-	-	-	7.8	10.7	>250	8.5	12.6	>250	9.4	9.6	50	6.8	6.4	10
41867	Dimethyl sulfone	3.0	2.9	>250	3.0	2.7	80	3.0	2.4	>250	3.0	3.3	40	2.9	2.6	>250
					7.9	3.8		7.8	2.5		7.9	3.4		7.9	5.6	
					6.7	2.4		6.7	1.4		6.7	2.5		6.7	3.7	
					4.3	2.8		4.2	1.9		4.3	3.3		4.3	4.1	
					3.1	2.0		3.0	1.5		3.0	2.2		3.0	3.6	
50409	Thymol	-	-	-	1.4	2.5	>250	1.3	2.1	>250	1.4	2.7	>250	1.3	3.6	>250
					7.1	3.8		7.0	2.0		7.0	3.7		7.1	4.9	
					6.8	4.5		6.7	2.2		6.6	4.4		6.7	5.7	
					6.6	4.8		6.5	2.6		6.5	5.8		6.6	5.7	
					3.2	4.3		3.1	2.3		3.2	4.0		3.2	5.2	
					2.3	3.1		2.2	2.0		2.2	2.8		2.2	3.5	
					1.3	1.9		1.1	0.9		1.2	1.8		1.2	2.4	
55177	Benzyl benzoate	-	-	-	5.4	4.3	>250	5.4	2.2	>250	5.4	2.0	>250	5.4	1.0	>250
					8.1	13.6		8.0	8.8		8.1	3.9		8.1	2.4	
74599	1,3,5-Trimethoxy-benzene	-	-	-	6.1	4.7	>250	6.1	3.2	>250	6.1	4.8	>250	6.7	3.4	>250
					3.8	2.2		3.7	1.4		3.8	2.7		4.3	2.6	
74658	1,2,4,5-Tetramethyl-benzene	-	-	-	7.0	6.1	>250	6.9	4.7	10	6.9	5.9	10	6.9	7.7	50
					2.2	4.0		2.1	2.6		2.2	4.3		2.2	5.0	
89151	Dimethylmalonic acid	1.3	1.0	100	-	-	-	1.3	0.7	250	1.4	1.0	250	1.4	2.0	30
92816	Maleic acid	6.3	6.1	>250	-	-	-	6.3	3.0	>250	6.3	3.9	10	6.4	2.3	20
94681	Methyl 3,5-dinitrobenzoate	-	-	-	9.3	8.0	>250	9.1	9.4	100	-	-	-	9.1	9.0	>250
					9.2	6.1		8.9	7.6		-	-		9.0	8.2	
					4.1	2.6		4.0	1.5		-	-		4.0	3.5	

NMR in Clinical Metabolomics : Ivdr

Compound	Conc. mmol/L	LOD mmol/L	r mmol/L	p %	Δ mmol/L	95% Range mmol/L	Graphics (*)
2-Aminobutyric acid	< 0.05	0.05	0.031	8 ○	0.087	≤ 0.10	
Alanine	0.35	0.02	0.351	100 ●	0.010	0.29 - 0.64	
Asparagine	< 0.05	0.05	0.000	0 ○	5.082	≤ 0.08	
Creatine	0.01	0.01	0.014	100 ●	0.002	≤ 0.07	
Creatinine	0.09	0.01	0.091	100 ●	0.002	0.06 - 0.14	
Glutamic acid	0.10	0.05	0.099	87 ○	0.025	≤ 0.24	
Glutamine	0.83	0.02	0.832	99 ●	0.023	0.30 - 0.83	
Glycine	0.26	0.01	0.257	100 ●	0.006	0.17 - 0.44	
Histidine	0.13	0.02	0.127	100 ●	0.002	0.07 - 0.16	
Isoleucine	0.04	0.03	0.043	93 ○	0.010	0.03 - 0.11	
Leucine	0.09	0.01	0.090	95 ●	0.013	0.07 - 0.20	
Lysine	0.19	0.04	0.194	66 ○	0.073	≤ 0.29	
Methionine	0.09	0.05	0.093	97 ●	0.009	0.05 - 0.13	
N,N-Dimethylglycine	< 0.01	0.01	0.001	88 ○	0.000	≤ 0.01	
Ornithine	< 0.02	0.02	0.000	0 ○	2.497	≤ 0.16	
Phenylalanine	< 0.03	0.03	0.028	97 ●	0.003	≤ 0.07	
Proline	0.29	0.05	0.293	24 ○	3.267	≤ 0.59	
Sarcosine	< 0.01	0.01	0.004	80 ○	0.001	≤ 0.01	
Threonine	< 0.04	0.04	0.000	0 ○	3.053	≤ 0.24	
Tyrosine	0.04	0.03	0.038	97 ●	0.003	≤ 0.08	
Valine	0.23	0.03	0.233	100 ●	0.005	0.15 - 0.35	

(*) Gray horizontal boxes represent 95% concentration range, black vertical lines represent sample value.

Bruker 600 MHz AVANCE™ IVDr



- Automatized quantification of more than 40 metabolites in blood
- Absolut Quantification of metabolites (B.I.Methods™)
- Validation of LODs following the ISO 17025 guidelines.
- Exploration and quantification of a huge panel of lipoproteins

NMR in Clinical Metabolomics : large scale studies

Robustness

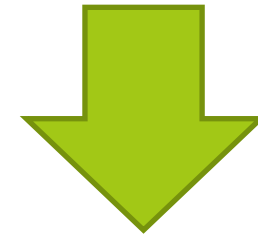
Reproducibility

Quantification

Easy samples
preparation

poor time
consuming

Multi-sites comparison
Multi-datasets comparison



Large scale epidemiology
Longitudinal studies

NMR in Clinical Metabolomics : large scale studies

A Metabolic Profile of All-Cause Mortality Risk Identified in an Observational Study of 44,168 Individuals.
Deelen & al. 2019, Nat. Comm. 10, 3346

Aim: longer-term mortality risk prediction

44000 individuals coming from 12 cohorts

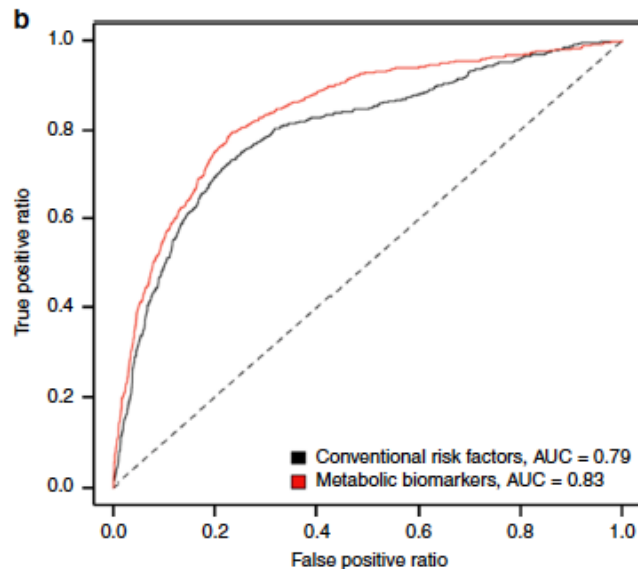
5500 dead during the follow-up process (2 to 16 years)



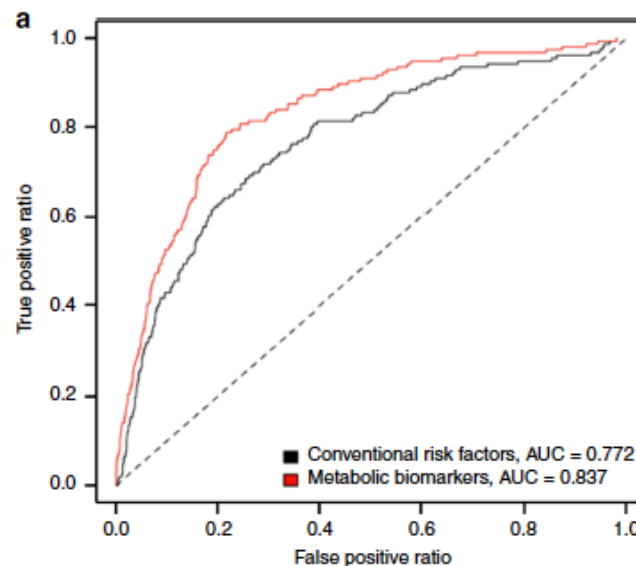
Serum
High-Troughput 1D NMR analysis
256 spectral zones quantified



14 identified biomarkers
associated with all cause
mortality after adjustment
for multiple testing



10 years



5 years

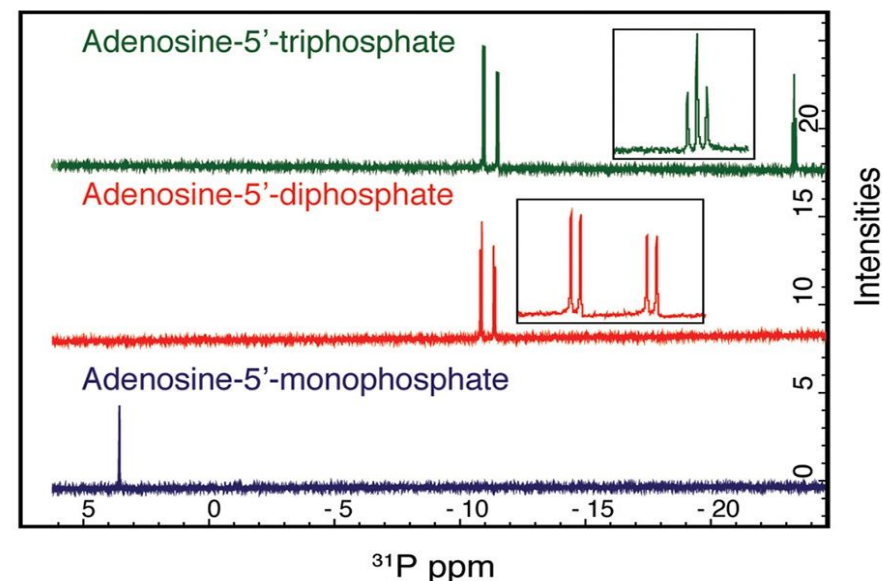
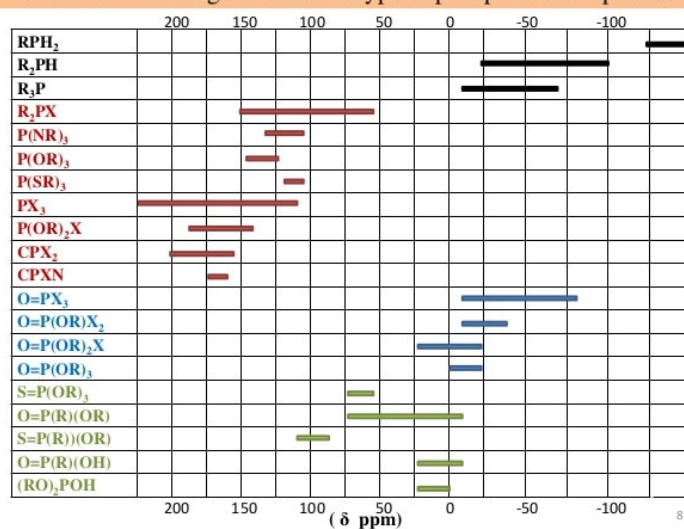
NMR in Clinical Metabolomics : ^{31}P spectroscopy

Phosphorylation and dephosphorylation of enzymes and proteins play a crucial role in many cellular processes

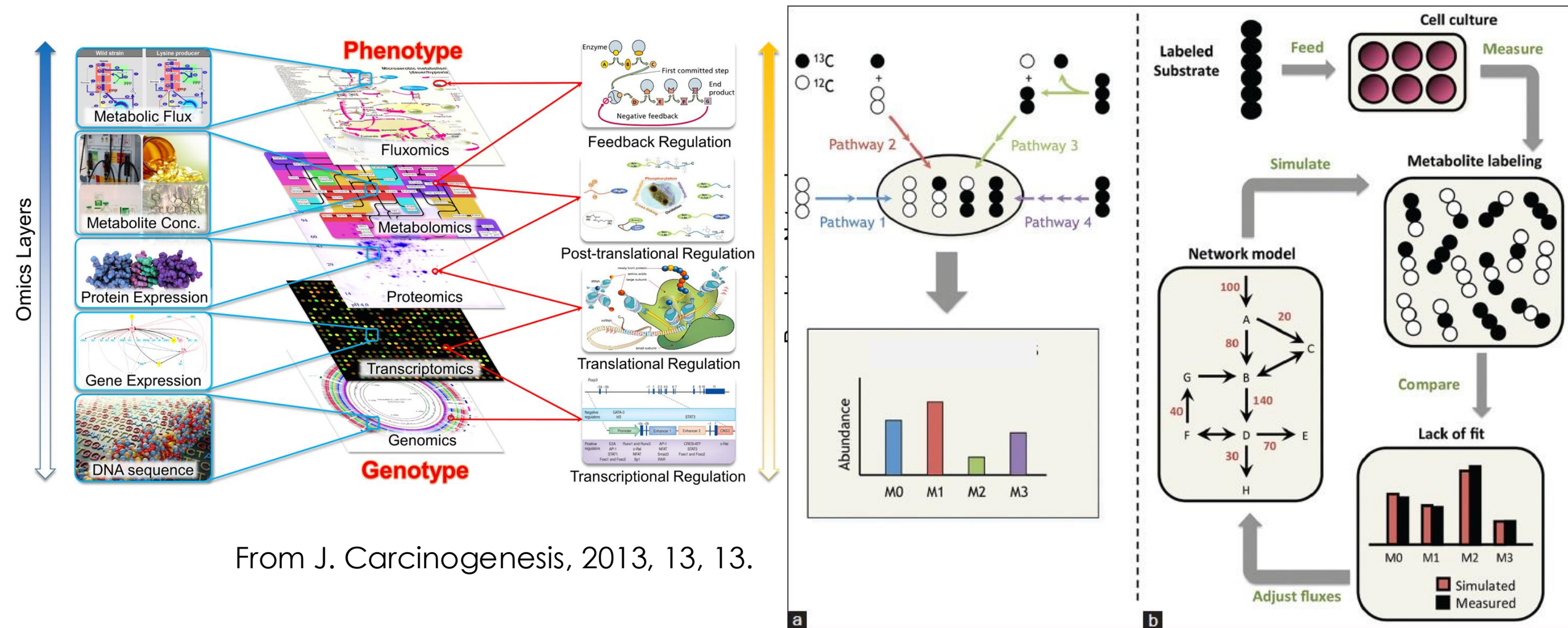
Many phosphorylated metabolites (NAD, NADH, NADP, UTP, CTP, ATP, ADP, AMP, ...) are the mainstays of the energy machinery of the cells

Phosphometabolites represent more than 30 % of identified metabolites

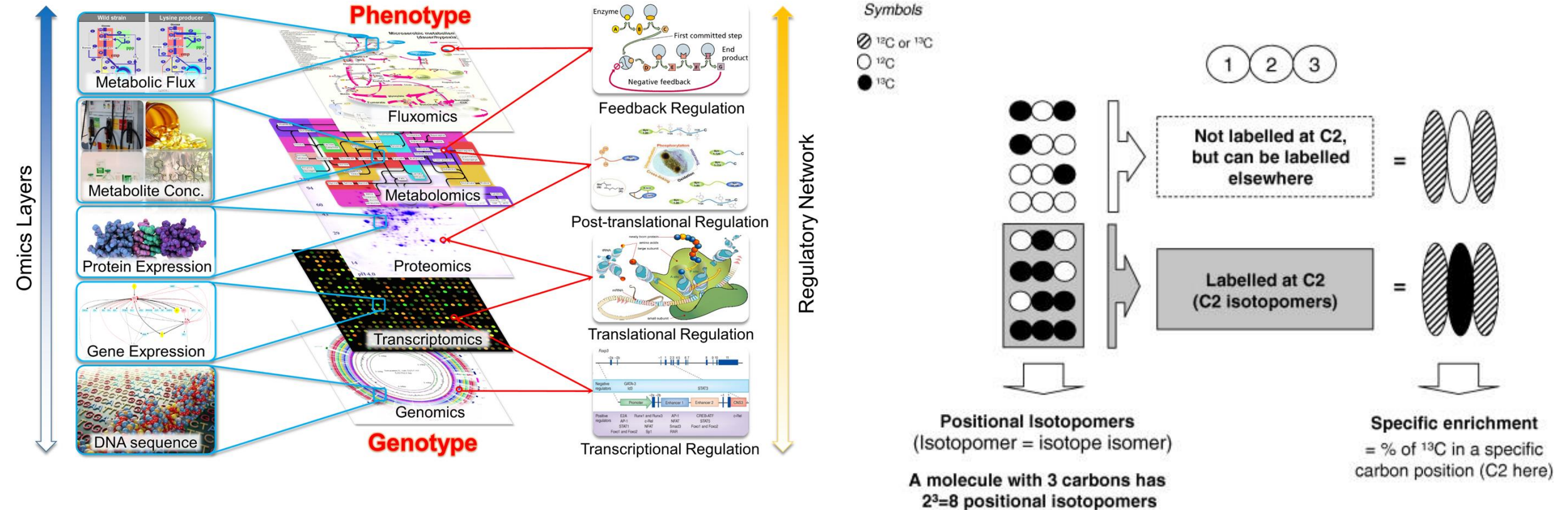
Chemical shift range of different type of phosphorus compounds



NMR in Clinical Metabolomics : ^{13}C spectroscopy and fluxomics



NMR in Clinical Metabolomics : ^{13}C spectroscopy and fluxomics



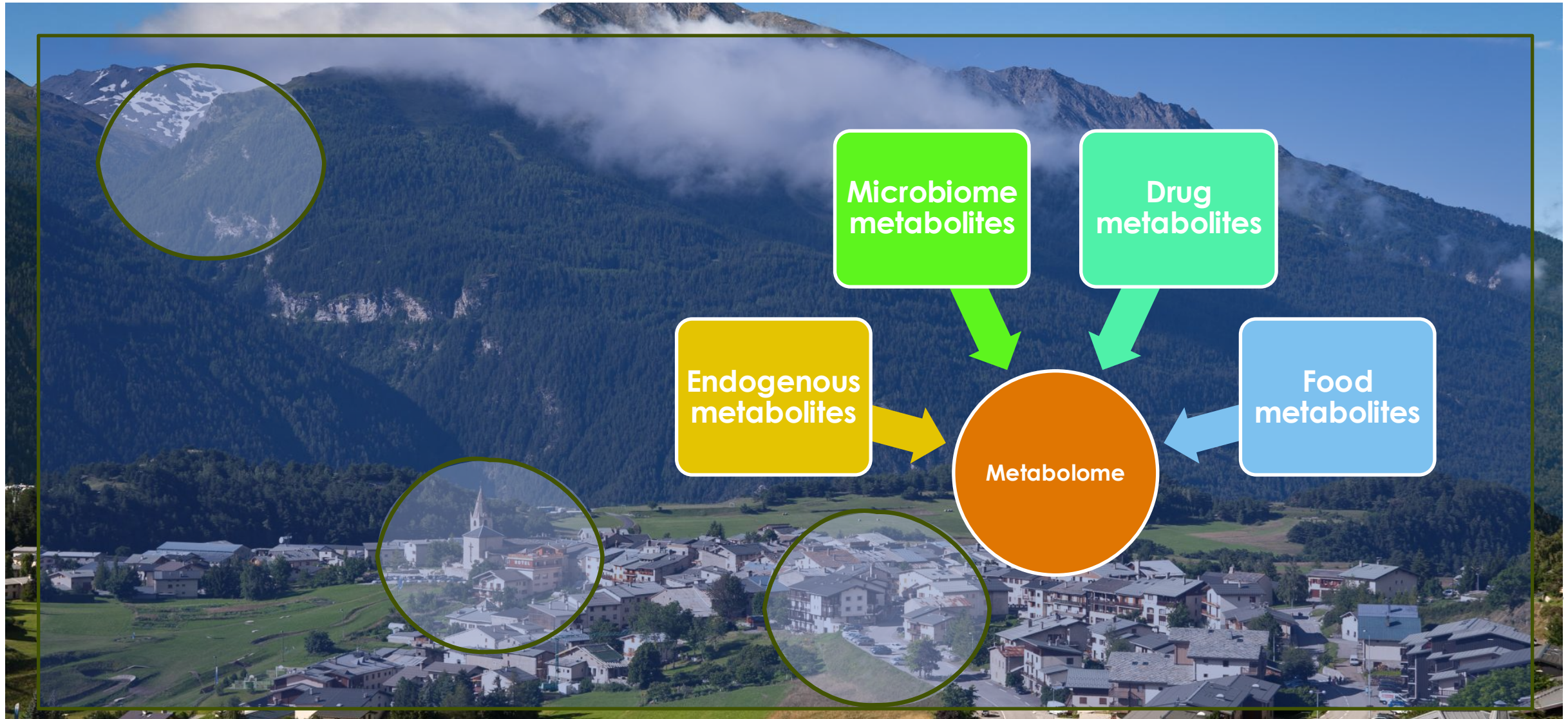
From J. Carcinogenesis, 2013, 13, 13.

NMR could help to identify the isotopomers

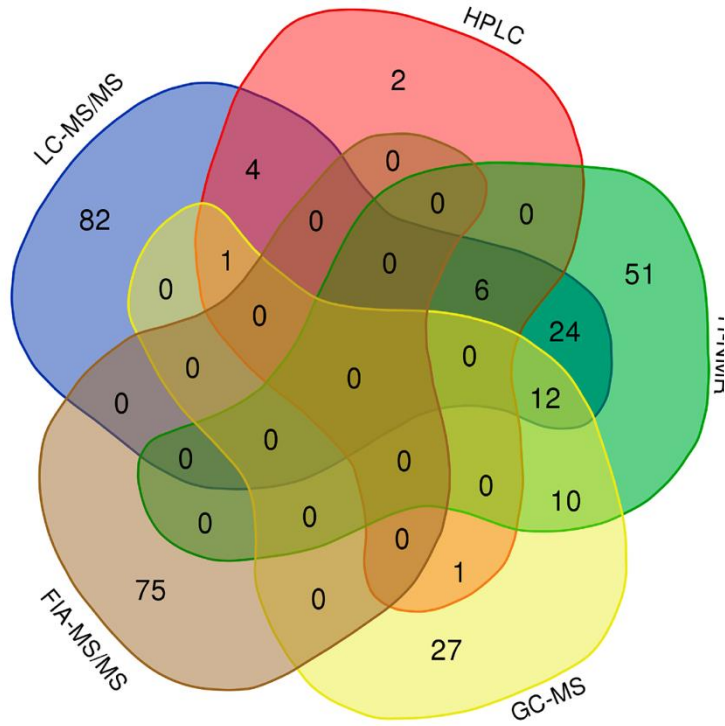
NMR and MS: complementary point of view



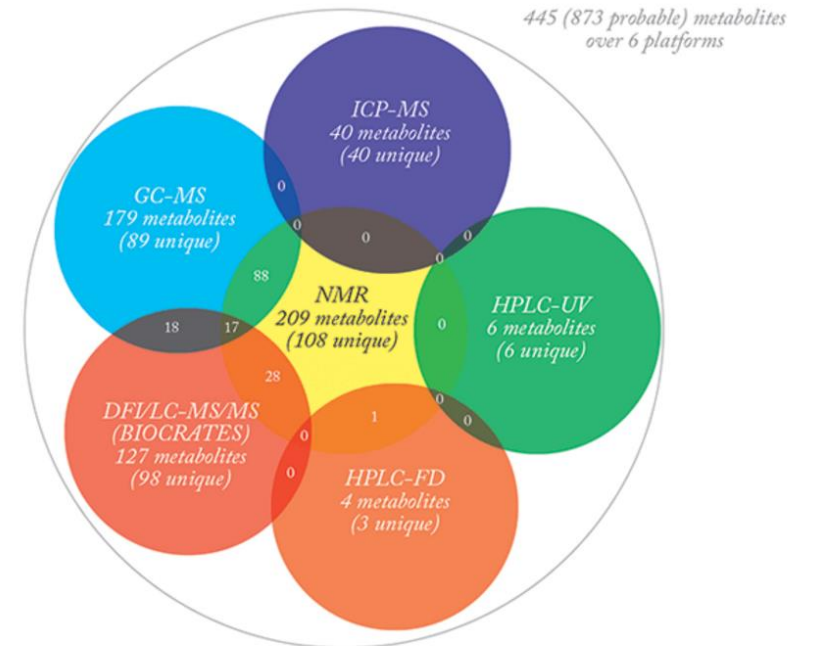
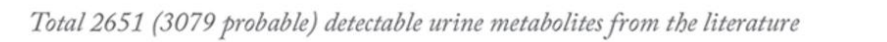
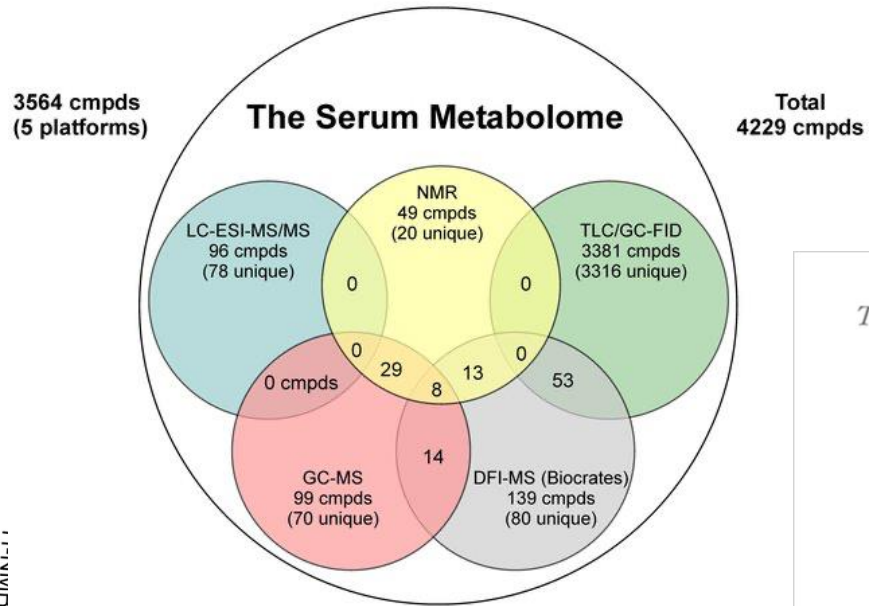
NMR and MS: complementary point of view



NMR and MS: increasing the metabolites coverage



Blood



445 (873 probable) metabolites
over 6 platforms

NMR and MS: increasing the metabolites coverage

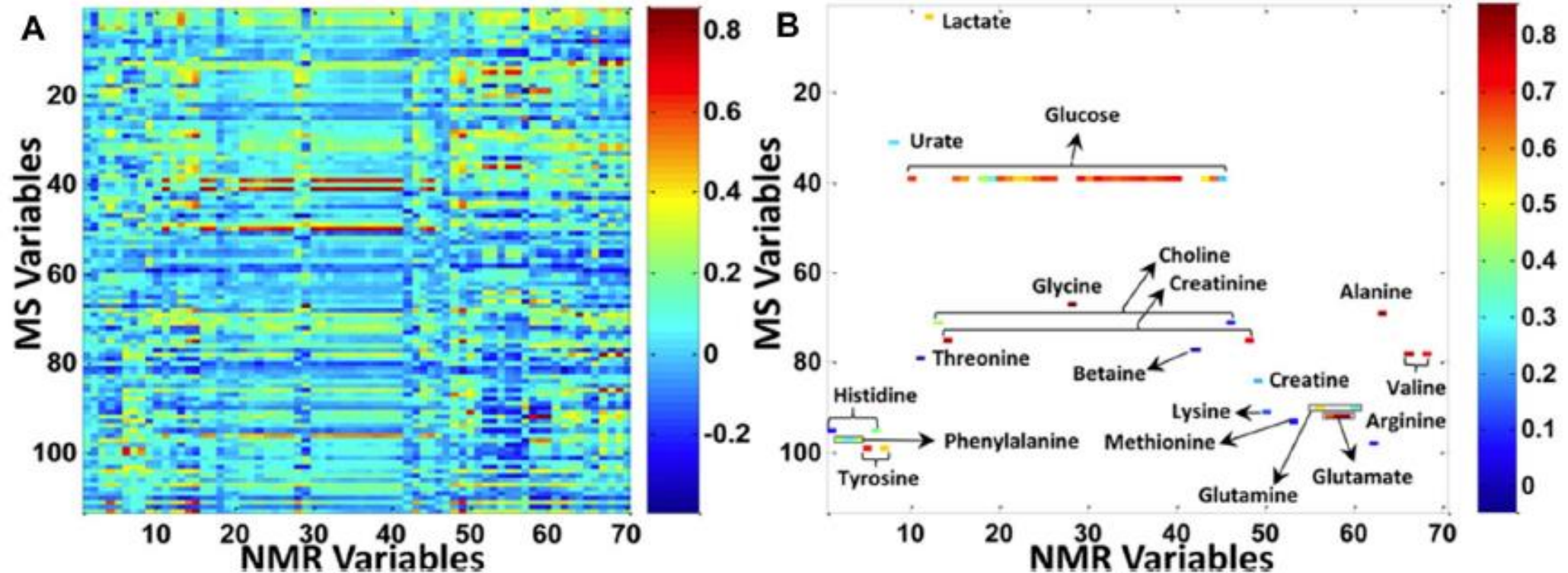


FIGURE 4 | Increasing the metabolic coverage of serum samples from patients suffering from colorectal cancer and polyps, by using both NMR and MS analytical platforms. **(A)** Correlation between all NMR and MS variables. **(B)** Correlation between the subset of metabolites (labeled in the figure) that can be detected by both NMR and MS. The X axis provides an index of all NMR variables in the data matrix, and the Y axis provides an index of all MS variables in the data matrix. Reproduced with permission from (Deng et al., 2016).

NMR and MS: improve the metabolites quantification

analytical
chemistry

Cite This: *Anal. Chem.* 2018, 90, 2001–2009

Article

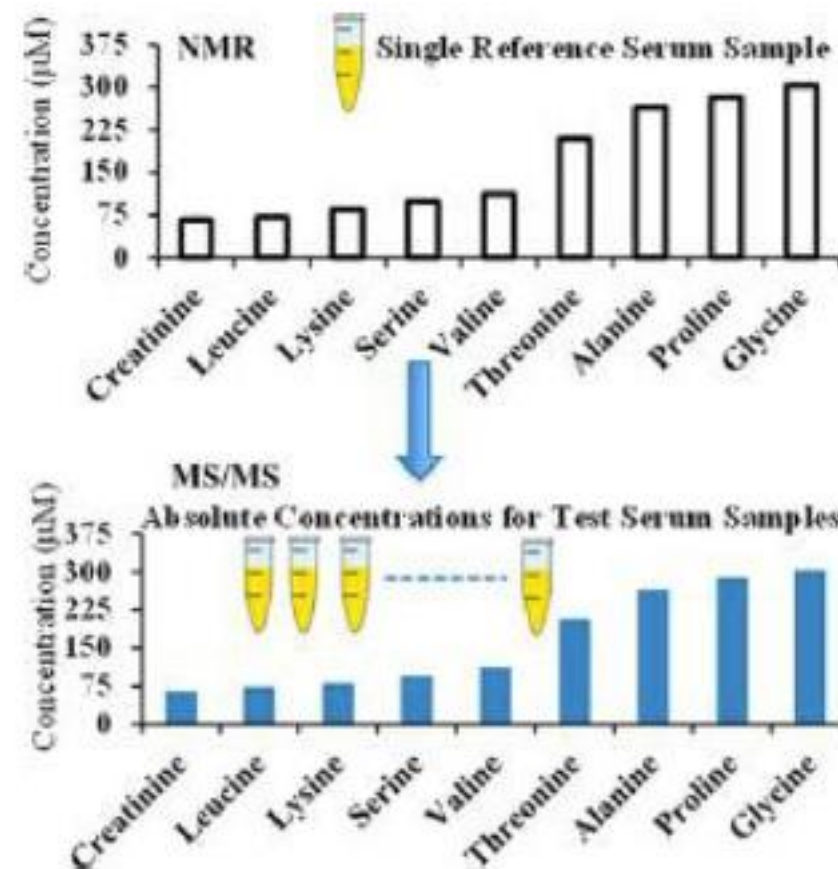
pubs.acs.org/ac

NMR-Guided Mass Spectrometry for Absolute Quantitation of Human Blood Metabolites

G. A. Nagana Gowda,^{*,†} Danijel Djukovic,[†] Lisa Fan Bettcher,[†] Haiwei Gu,[†] and Daniel Raftery^{*,†,‡,§}

[†]Northwest Metabolomics Research Center, Anesthesiology and Pain Medicine, and [‡]Department of Chemistry, University of Washington, Seattle, Washington 98109, United States

[§]Fred Hutchinson Cancer Research Center, Seattle, Washington 98109, United States



NMR as quality control for MS: inside the samples

Rapid identification of outliers

Verification of the samples type

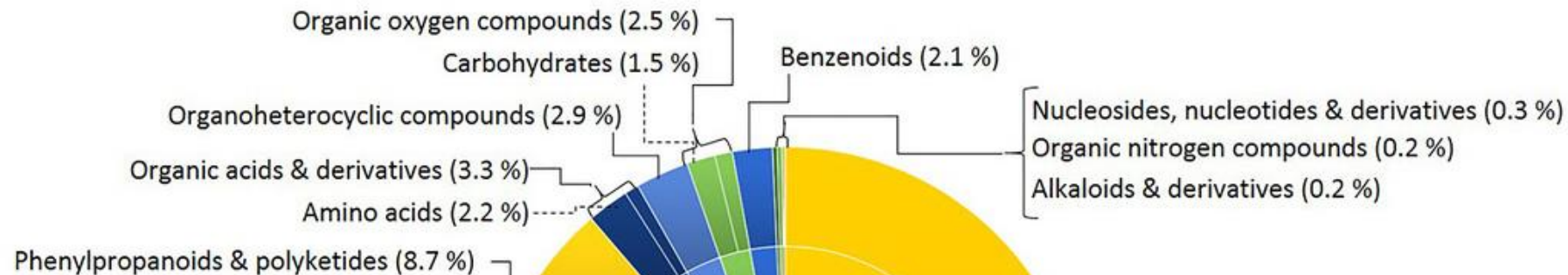
Verification of the quality of the samples

Samples selection for MS analysis

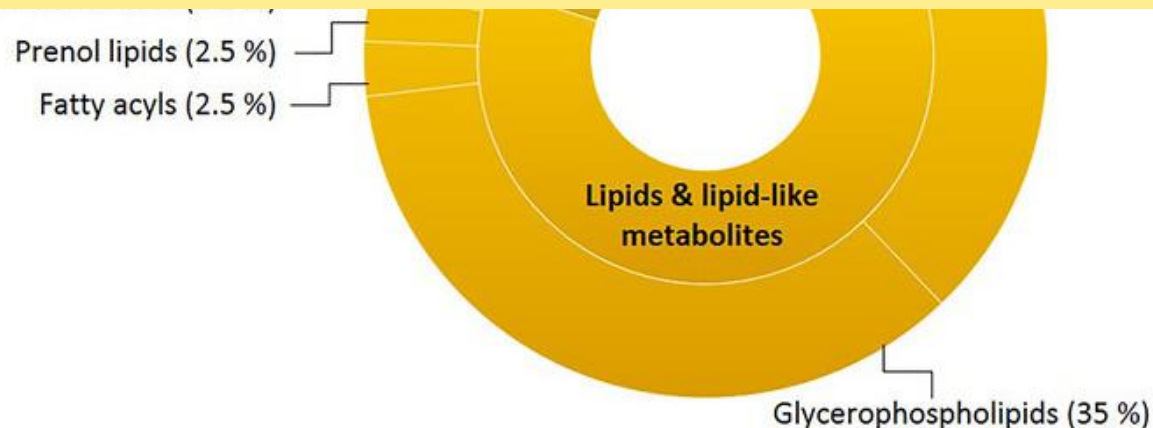
Selection of metabolites families to be explored by MS

NMR and MS complementarity MS in Clinical lipidomics

B) Chemical composition of human plasma metabolome



MS is more adapted to analyse the lipidome



Chemical classification

- Water-soluble polar metabolites
- Lipid and lipid-like non-polar metabolites

Biological roles

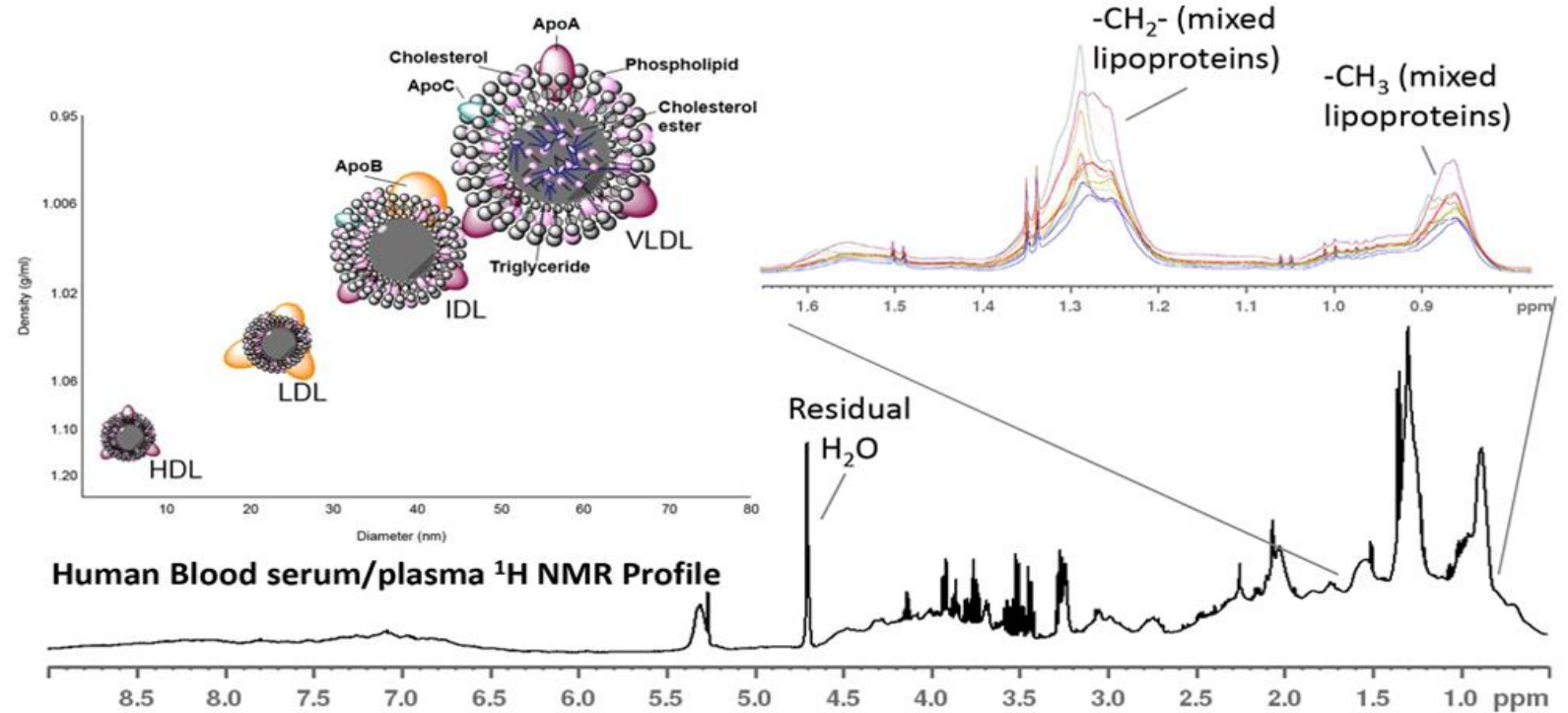
- Fuels (energy production and storage)
- Building blocks (cell structural integrity)
- Signaling molecules (signal transduction)

NMR and MS: the lipoproteins profile

- Lipoproteins = Carrier of Lipids in blood
- 4 main classes
- ≠ Biological properties
- ≠ Biochemical composition
- ≠ Physico-chemical properties

↓

≠ NMR spectral properties



Jimenez et al. 2018 Anal chem

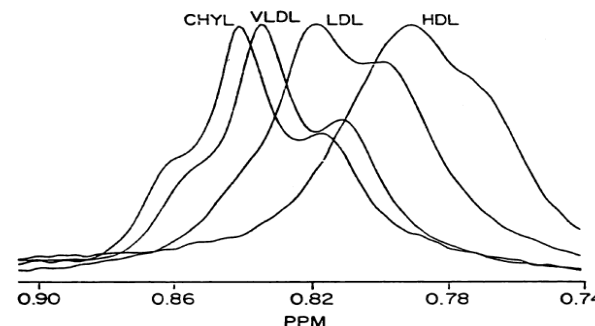
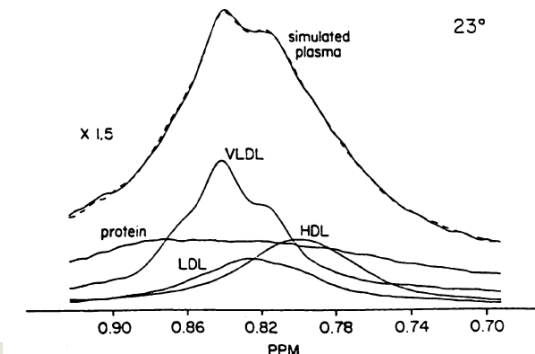
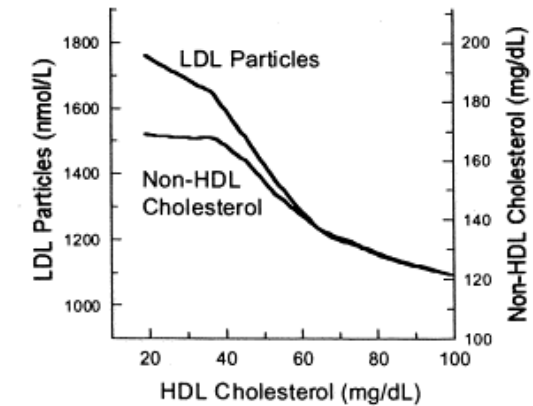
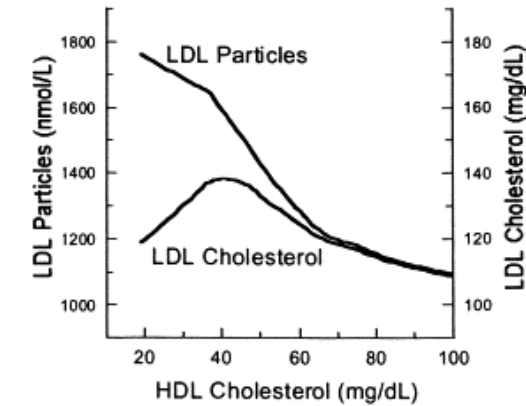
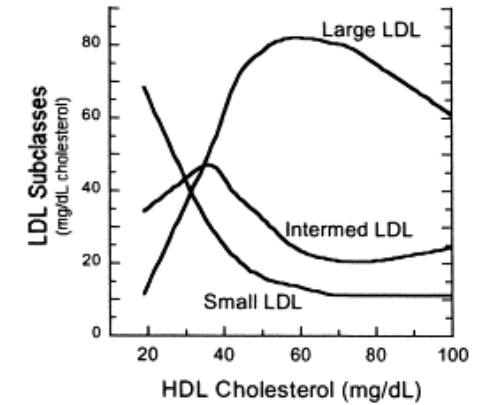
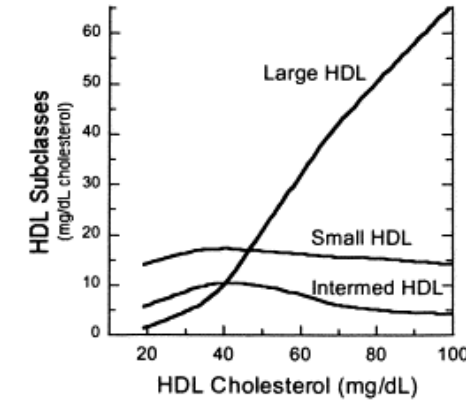
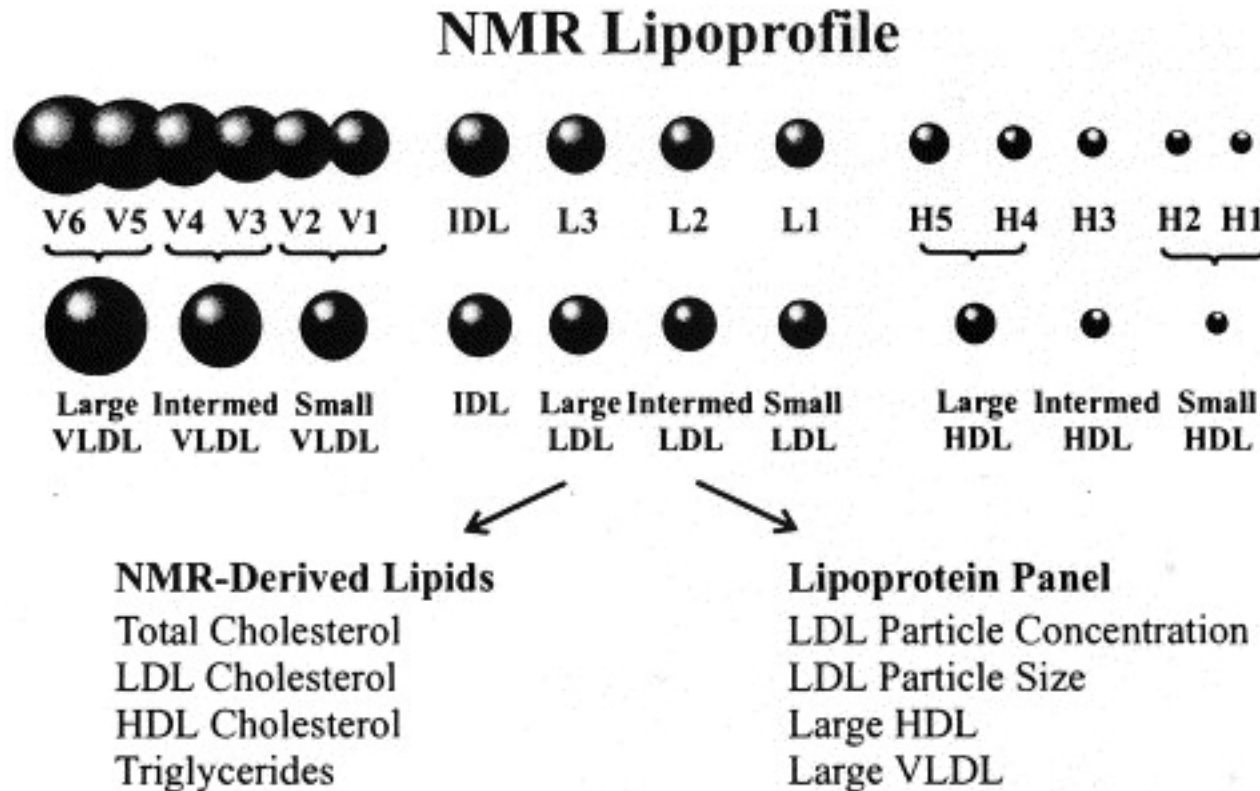


Fig. 1. Methyl lipid resonances of the major lipoprotein components of plasma [chylomicrons (CHYL), VLDL, LDL, HDL] obtained at 250 MHz and 45 °C
CLINICAL CHEMISTRY, Vol. 37, No. 3, 1991

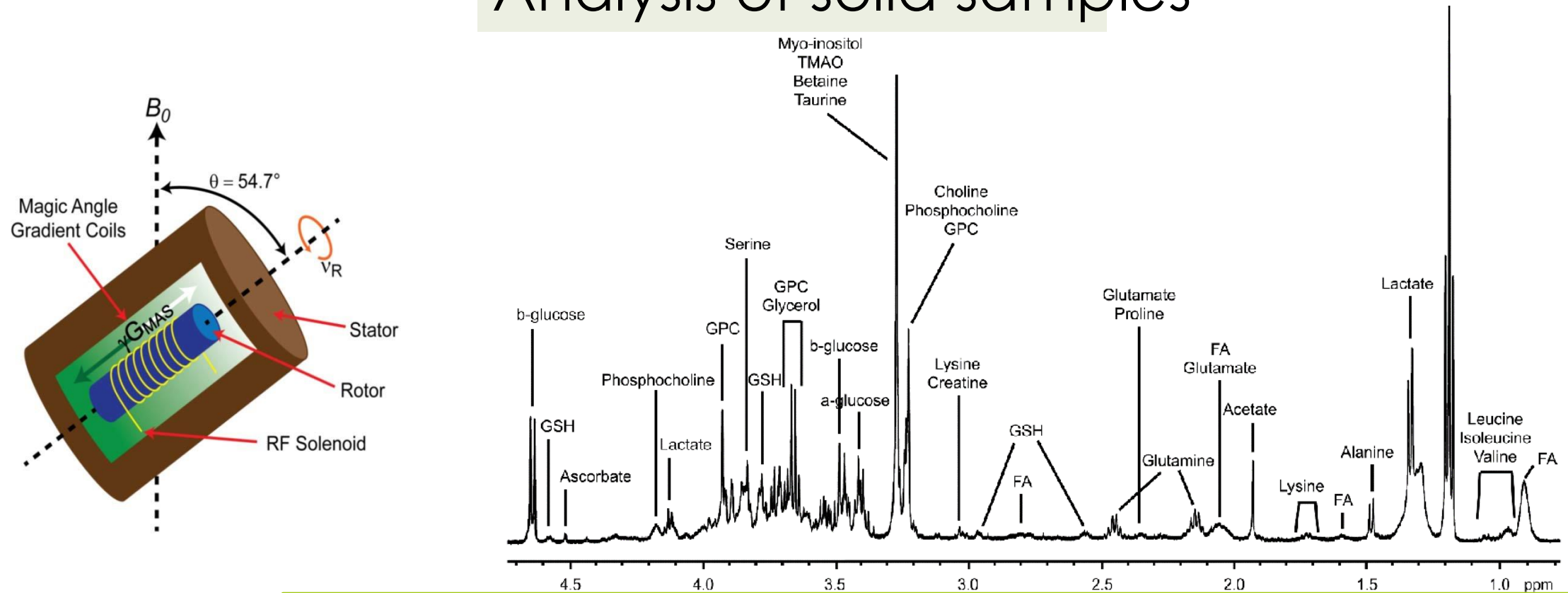


NMR and MS: the lipoproteins profile



HR-MAS

Analysis of solid samples



Representative 1D ^1H HRMAS NMR CPMG spectrum of liver tissue from a non-diabetic rat. Ethanol is a pollution resulting from the surgical material. FA: Fatty acids, GPC: Glycerophosphocholine, GSH: Glutathione, TMAO: Trimethylamine-N-oxide

From Vivot et al. Int. J. Biol. Sci. 2016

Pre-Analytic optimisation

Pre-analytics is essential

ISO 23118:2021

Standard offers technical specifications for metabolomics investigations in urine, serum, and plasma samples



**Many publications and recommendations
but mostly dedicated to biobank and/or research studies**

Pre-Analytic optimisations

Pulmonary revalidation study



1 to 4h



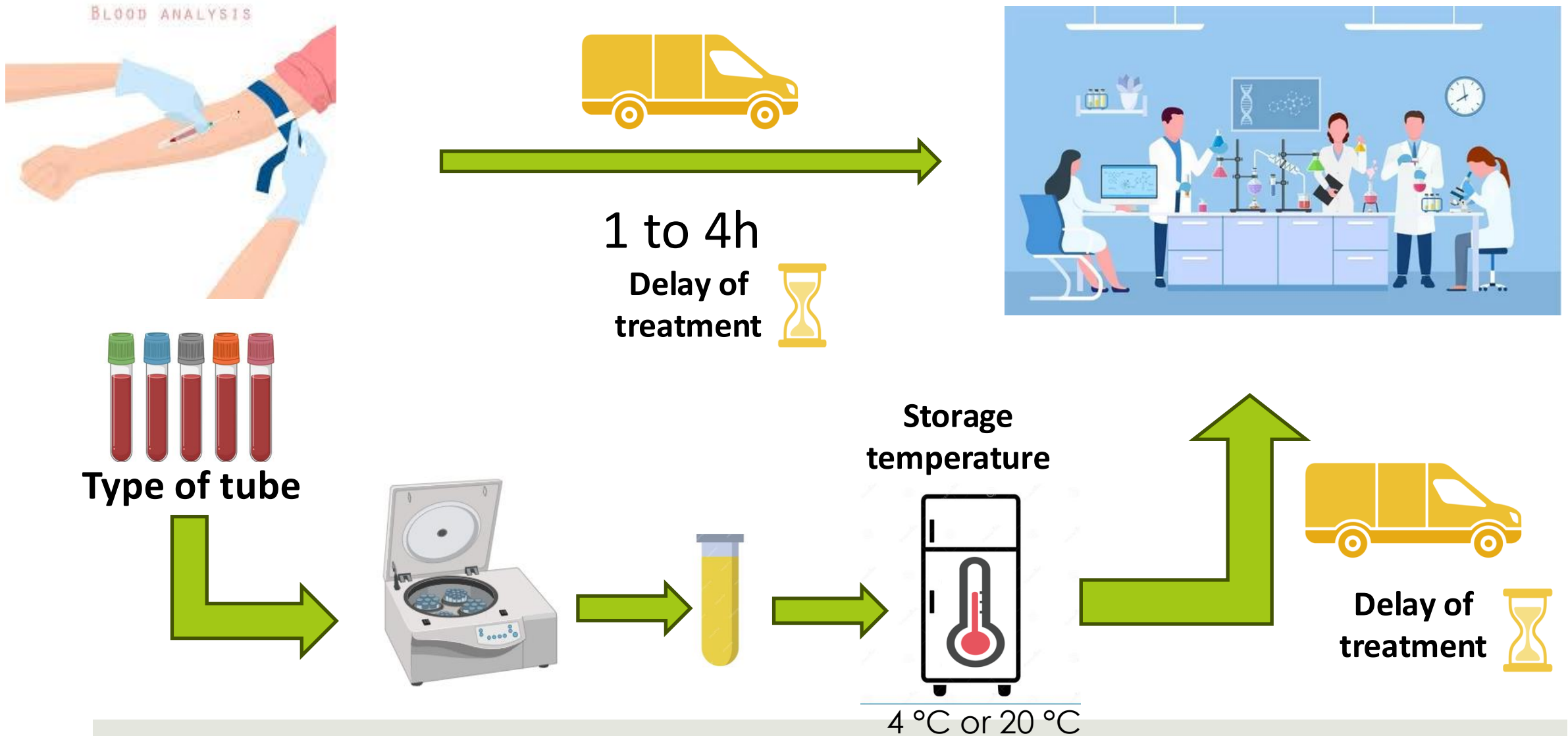
Collection center
Poor processing facilities

How to meet the quality standards
required in hospital laboratories ?

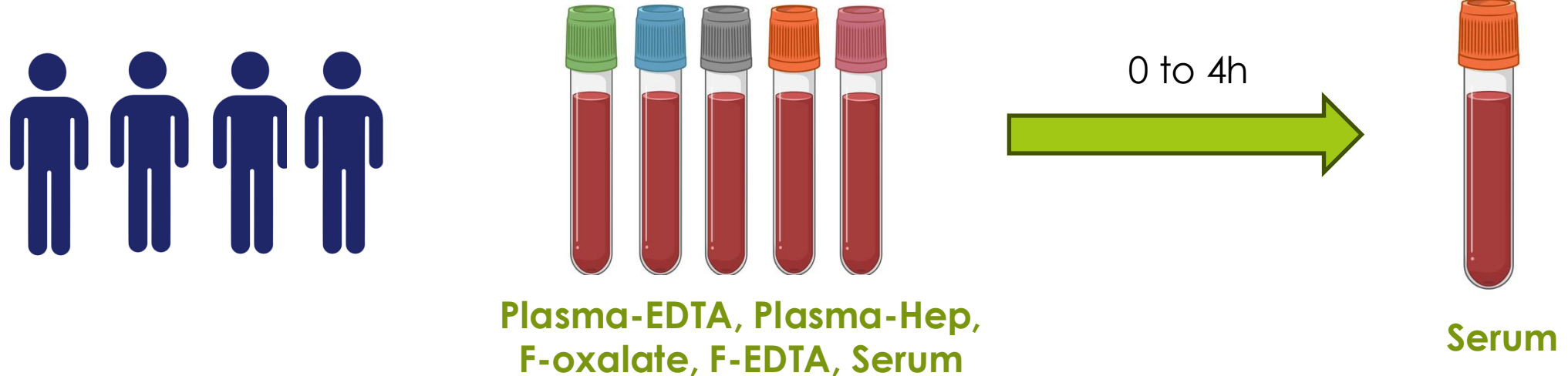
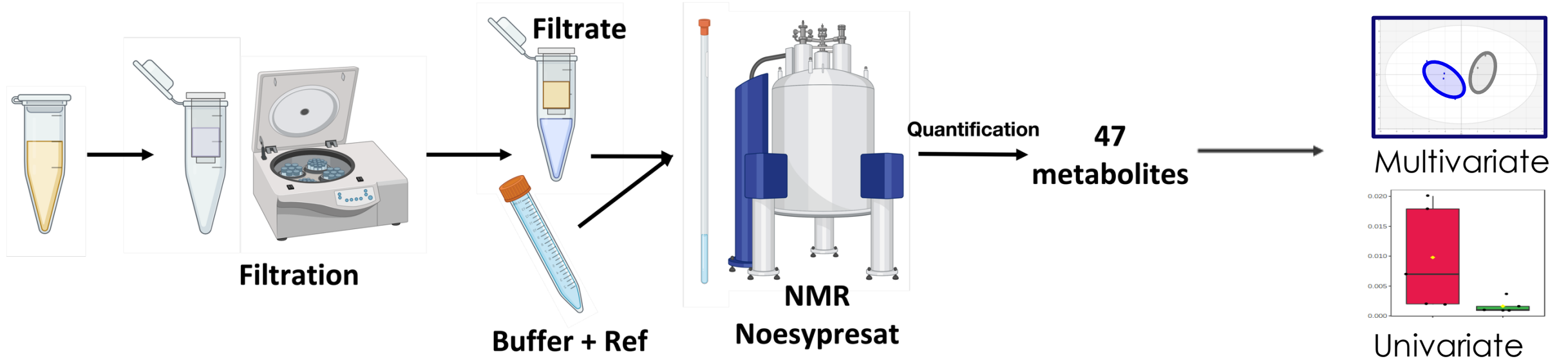


Can we export metabolomics out of
highly control environment?

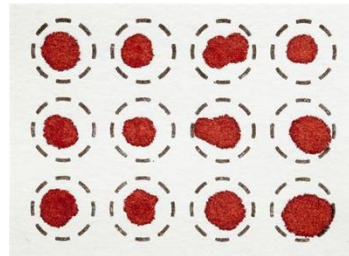
Pre-Analytics optimisations



Pre-Analytics optimisations



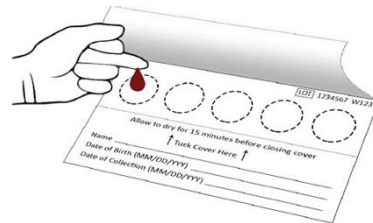
Exploration of new matrices



Sweat



Saliva



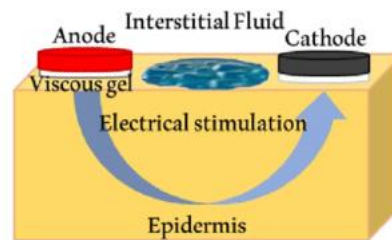
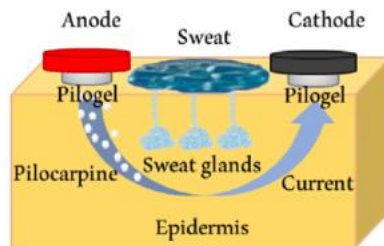
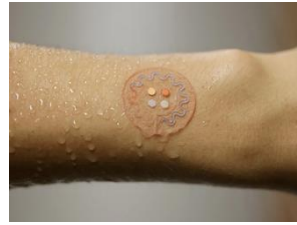
DBS



Stools

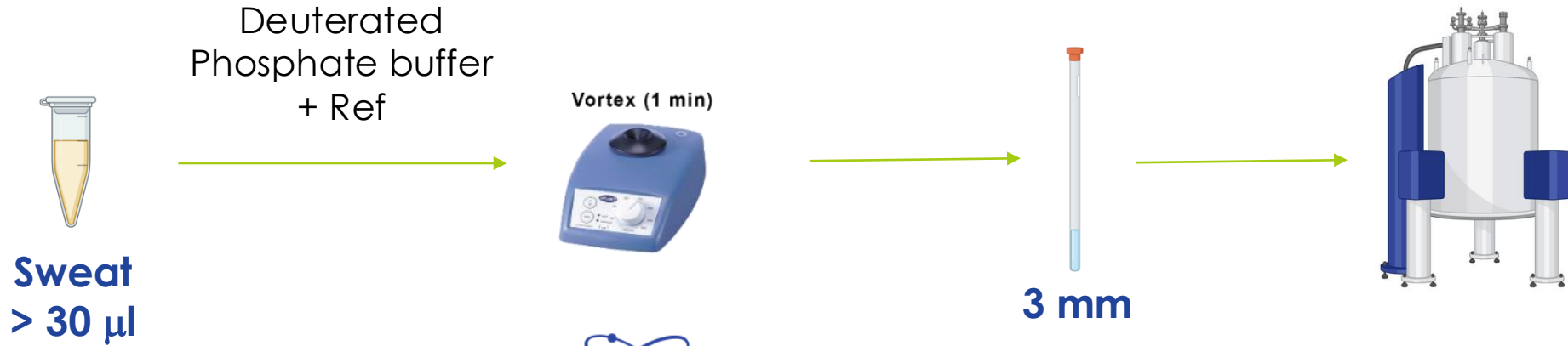


Preservation fluid

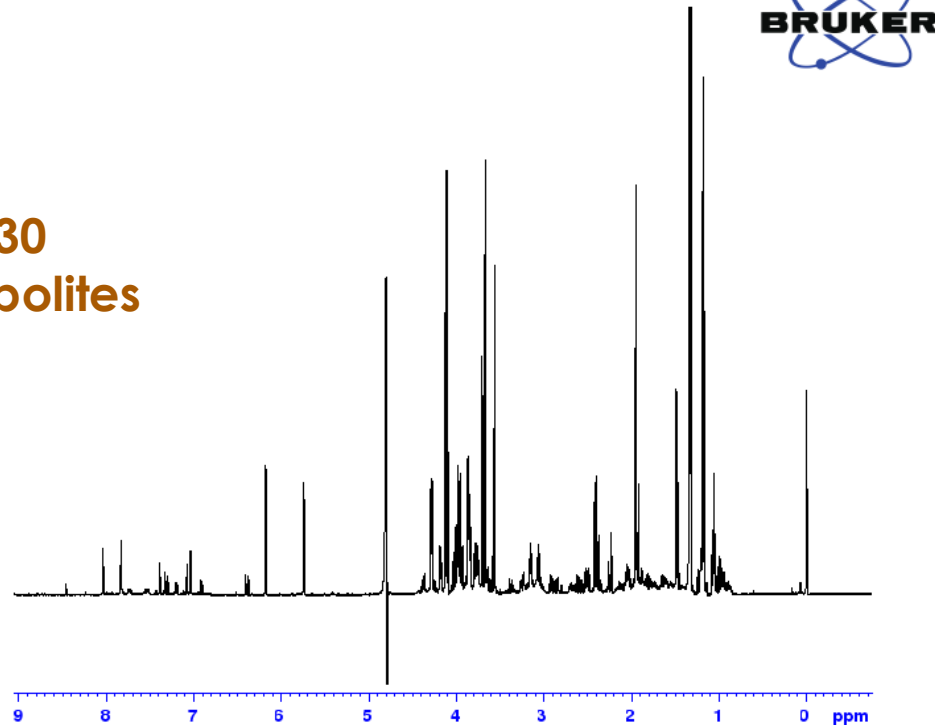


A vibrant collage of various sports-related items. At the top left is a round analog clock. Next to it is a green tennis ball. In the top center is a yellow star. To the right is a red boxing glove. Further right is a yellow baseball bat. At the top right is a white tennis ball. Below the clock is a white jump rope. To its right is a red tennis racket. Next to the racket is a grey bowling ball. To the right of the bowling ball is a white dumbbell. Further right is a light green t-shirt. Below the t-shirt is a pair of green-rimmed goggles. In the center, the word "Sport" is written in a large, bold, black font. To the left of "Sport" is a grey barbell. Below the barbell is a black and white soccer ball. To the right of the soccer ball is a yellow tennis racket. Further right is a red basketball. Below the basketball is a yellow dumbbell. At the bottom left is a yellow trophy cup. Next to it is a white bowling pin. To its right is a blue pair of shorts. Further right is a golf club. At the bottom right is a brown football. Various small stars and dots are scattered throughout the background.

Sweat spectral data



20 to 30
metabolites



Word cloud of identified metabolites:

- ISOLEUCINE
- GLYCINE
- FORMATE
- CITRULLINE
- ACETONE
- ACETATE
- ALANINE
- ETHANOL
- GLUTAMATE
- ISOPROPANOL
- PYROGLUTAMATE
- LACTATE
- PROPIONATE
- VALINE
- TYROSINE
- THREONINE
- ORNITHINE
- HISTIDINE
- SÉRINE
- LEUCINE
- UROCANATE

Sweat : drawbacks



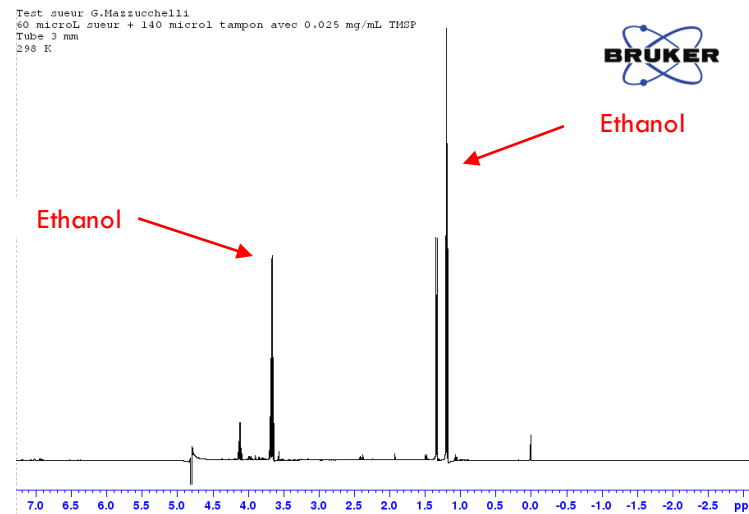
Contamination



Device or sample collection
protocol



Ethanol
isopropanol



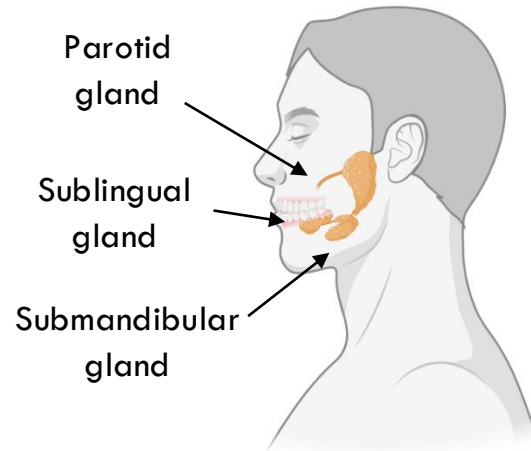
Normalization ?



SpeedVac Medium



Why Saliva ?



Rarely used compared to blood and urine...yet sampling :

- Simple
- Non invasive
- Fast collection
- Self-sampling



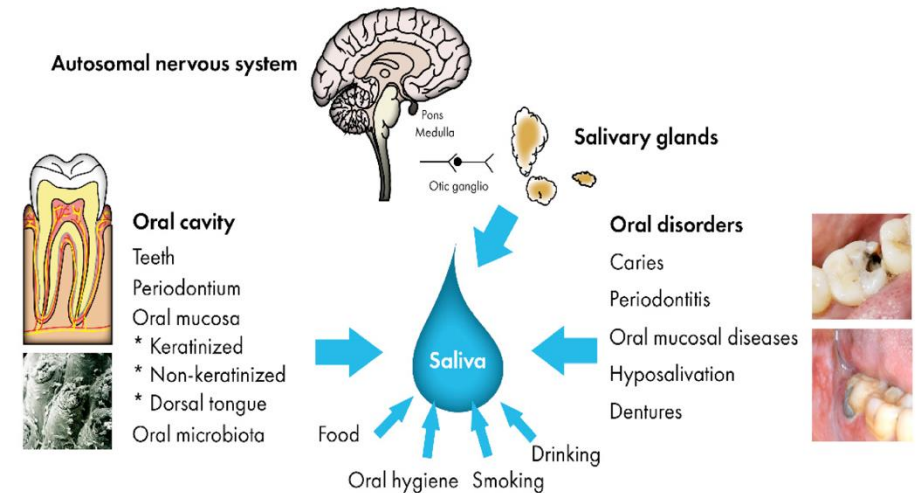
1 to 2 ml



Biomarkers

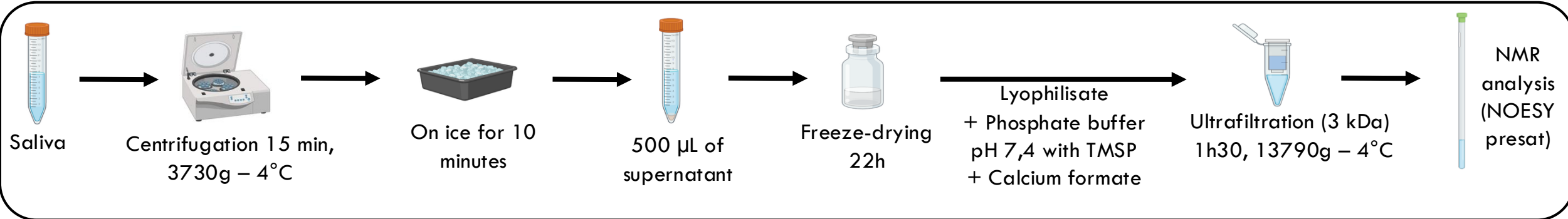


Sportomics



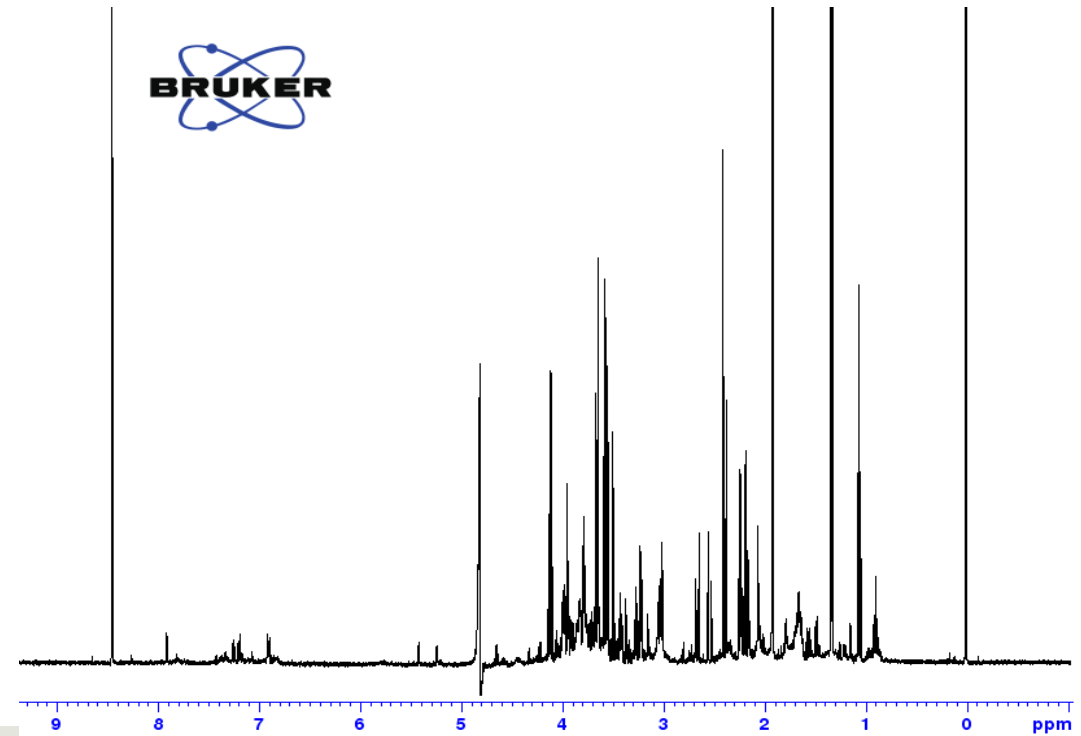
Oral diseases
Neck and oral cancer

Exemple de Saliva NMR spectrum



> 30 metabolites

ISOVALERATE
ISOCAPROATE
GLUCOSE
ETHANOL
PHENYLACETATE
CHOLINE
LACTATE
ACETATE
FUCOSE
CITRATE
ETHANOLAMINE
HISTIDINE
ISOLEUCINE
PHOSPHOÉTHANOLAMINE
TAURINE
GLYCINE
PROPIONATE
VALINE
PYRUVATE
SUCROSE
METHANOL
PHENYLALANINE
PUTRESCINE
SUCCINATE
UREA

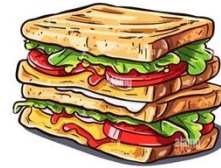


Saliva: drawbacks



ATTENTION

Contamination

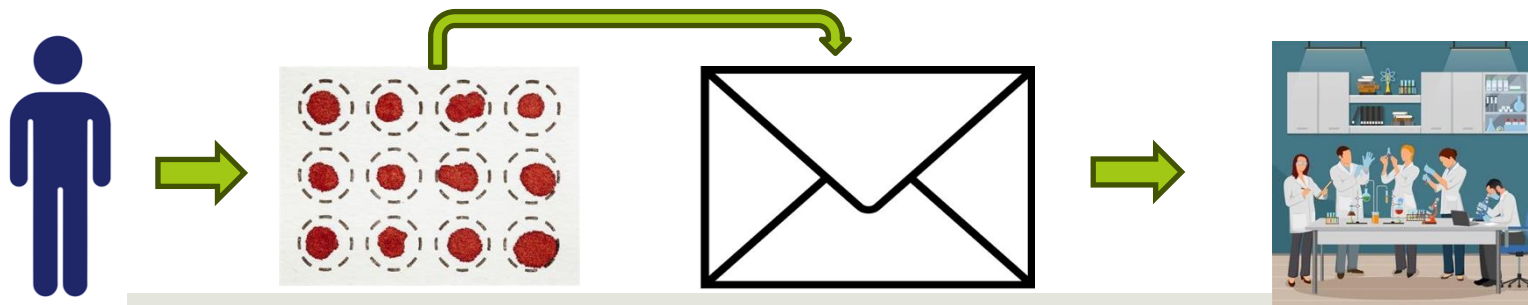
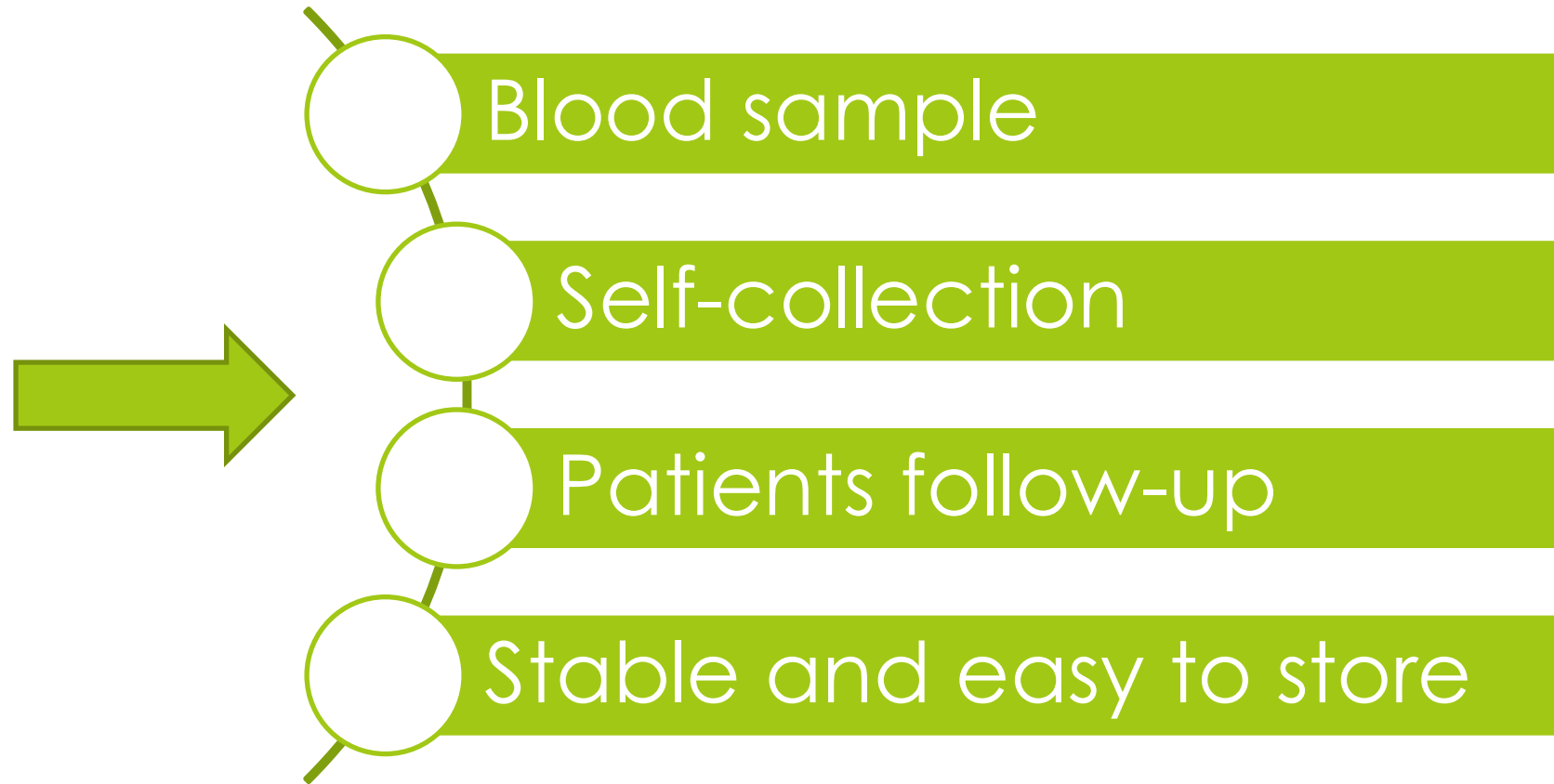
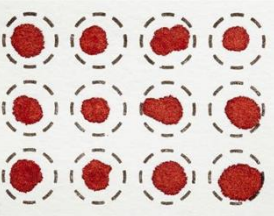


**Controlled protocol
for collection**

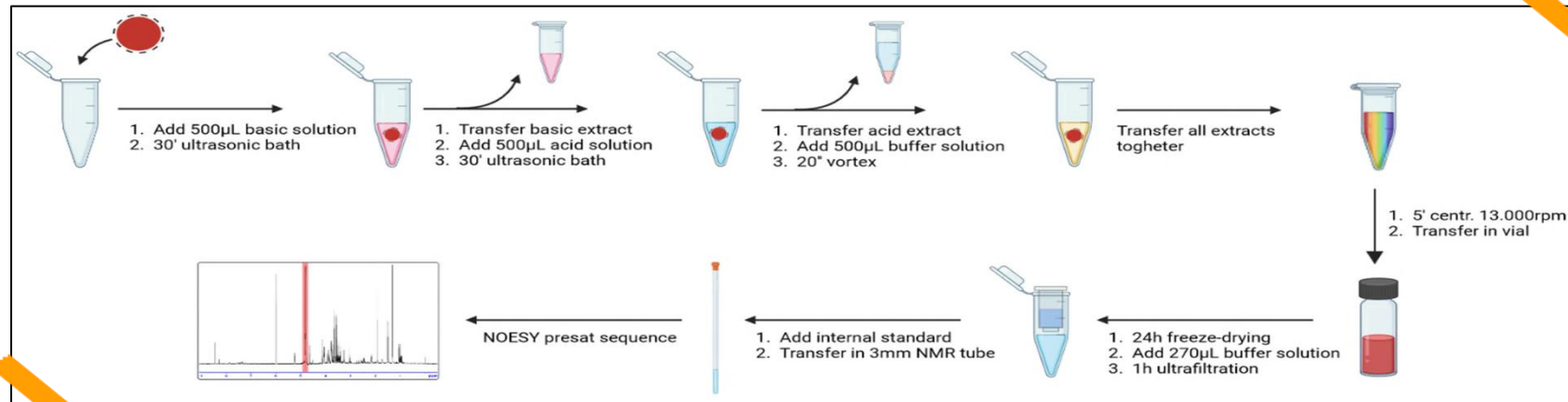
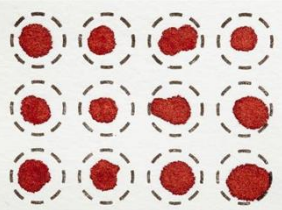


Normalization ?

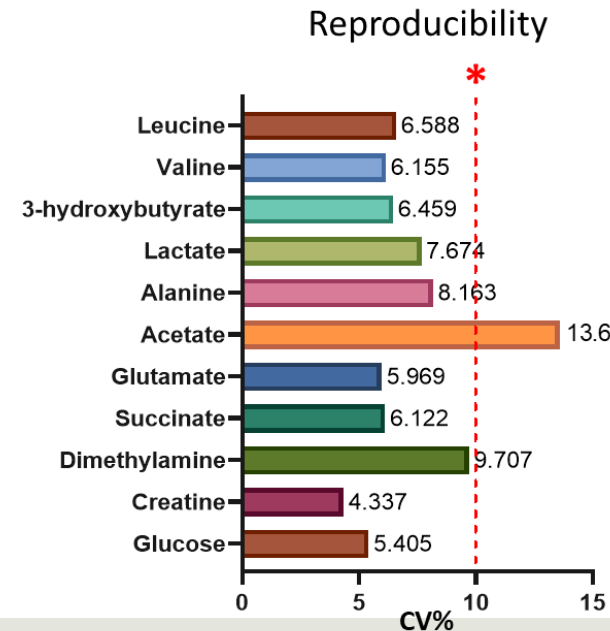
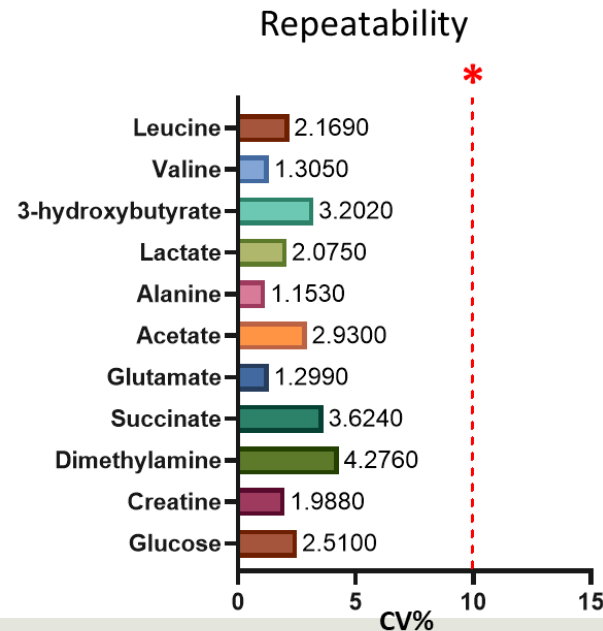
Why DBS ?



DBS: the extraction protocol

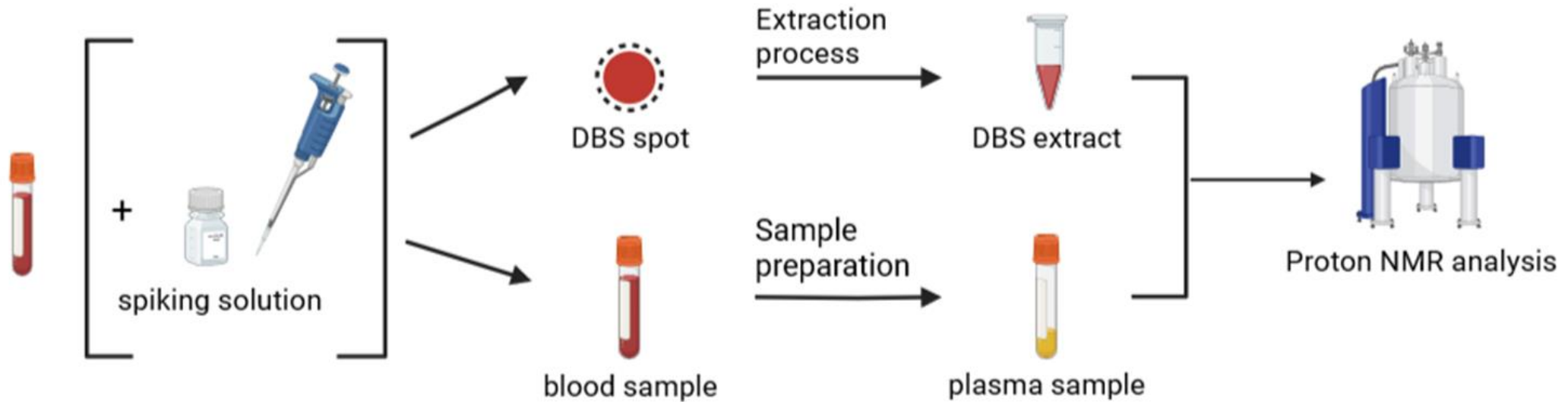
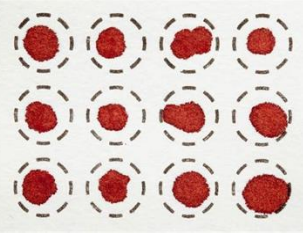


5 spots from the same individuals



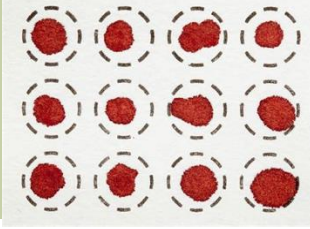
15 spots from the same Individuals Analyzed at D1, D2 and D3

DBS: comparison with plasma

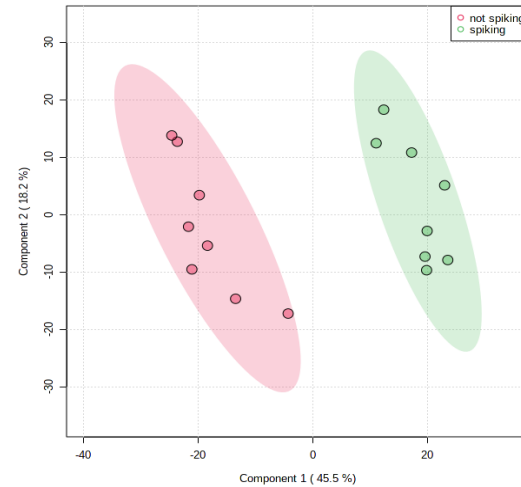


**Spiking solution = mixture of 10 metabolites
at a concentration close to 3 times the natural abundance in plasma**

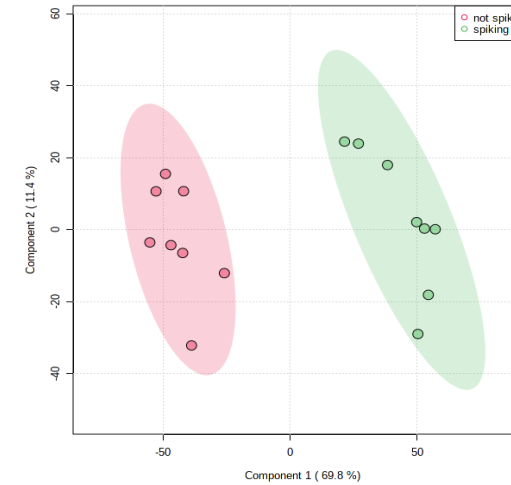
DBS: comparison with plasma



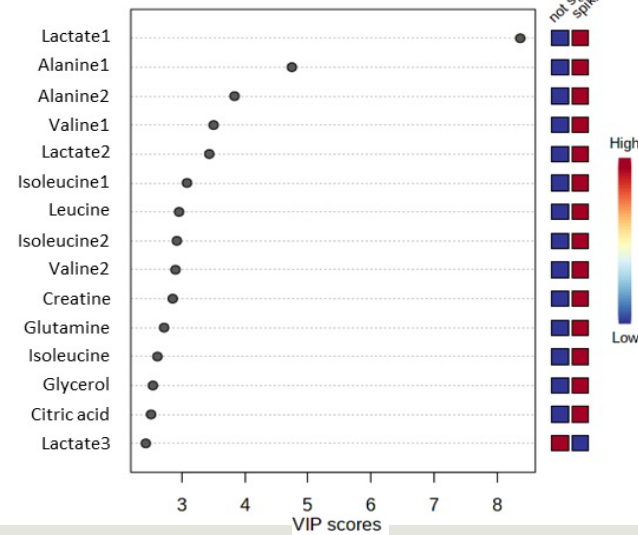
a PLS-DA: DBS



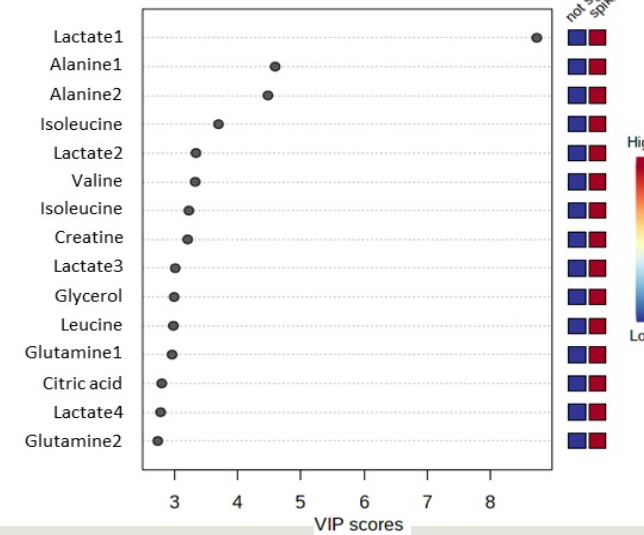
b PLS-DA: plasma



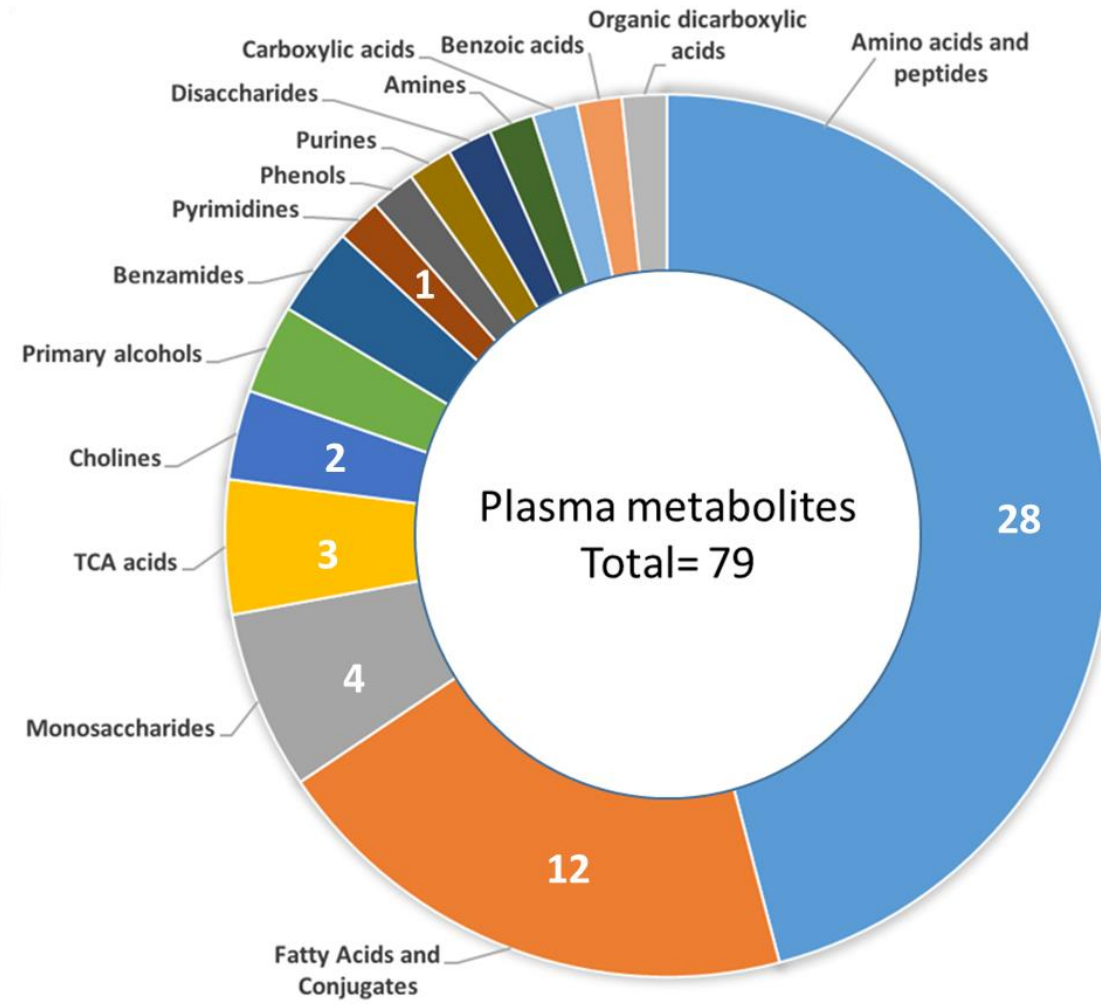
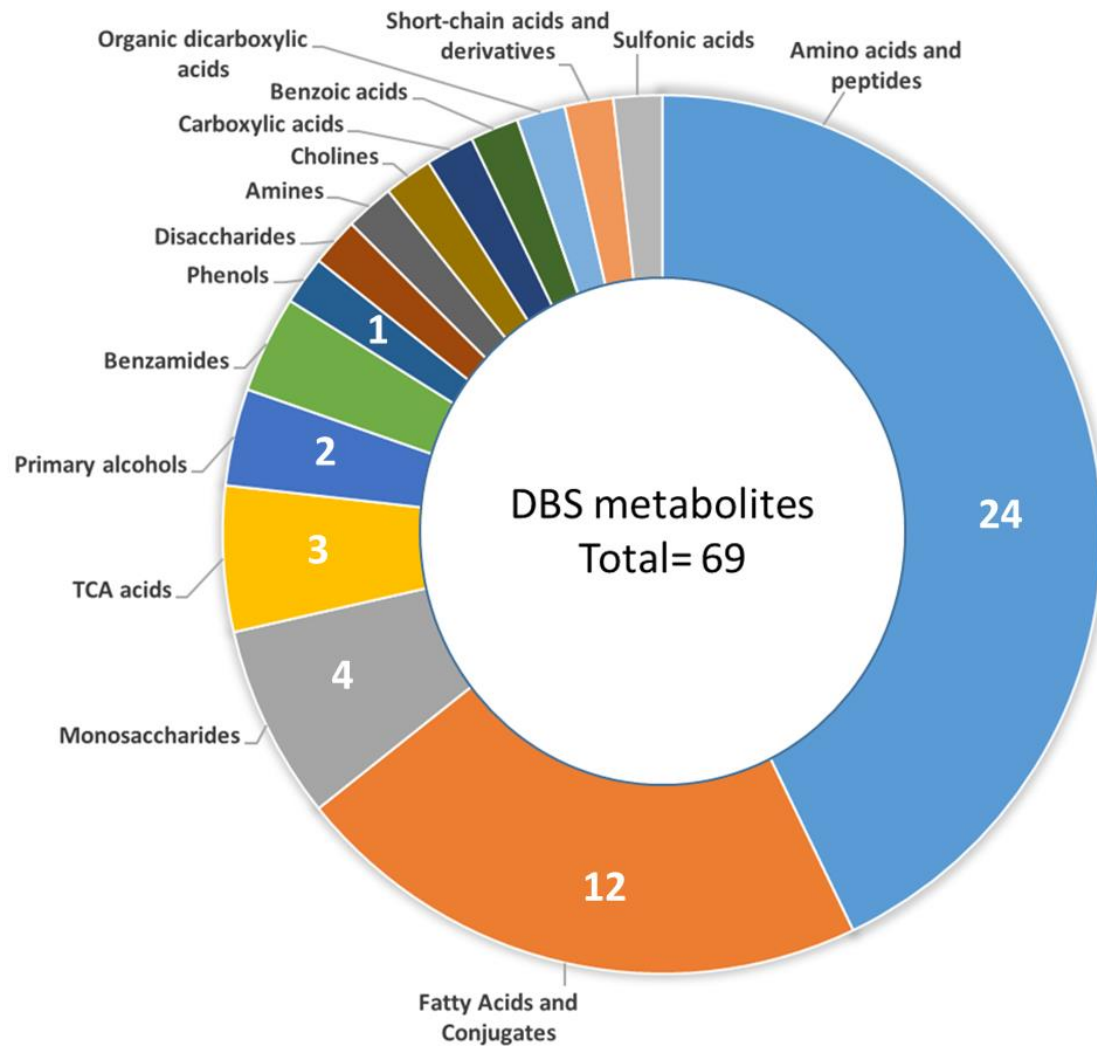
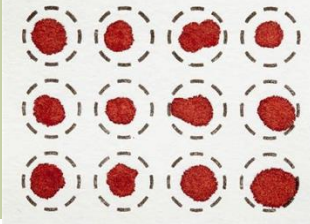
c VIP scores: DBS



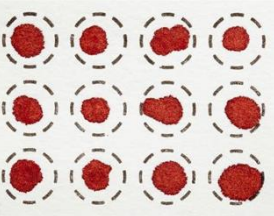
d VIP scores: plasma



DBS: comparison with plasma



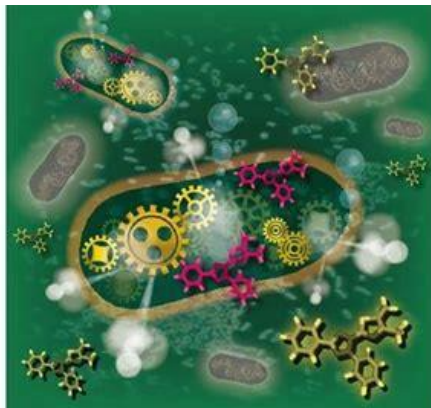
DBS: the drawback



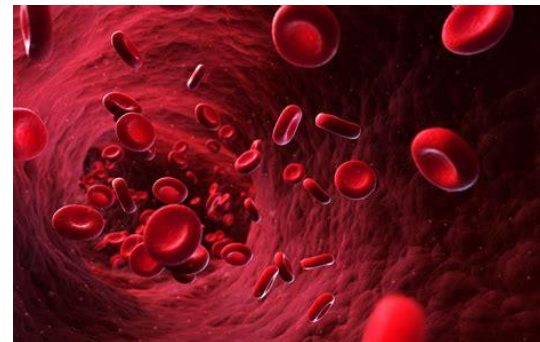
Huge contamination



Wash before collection



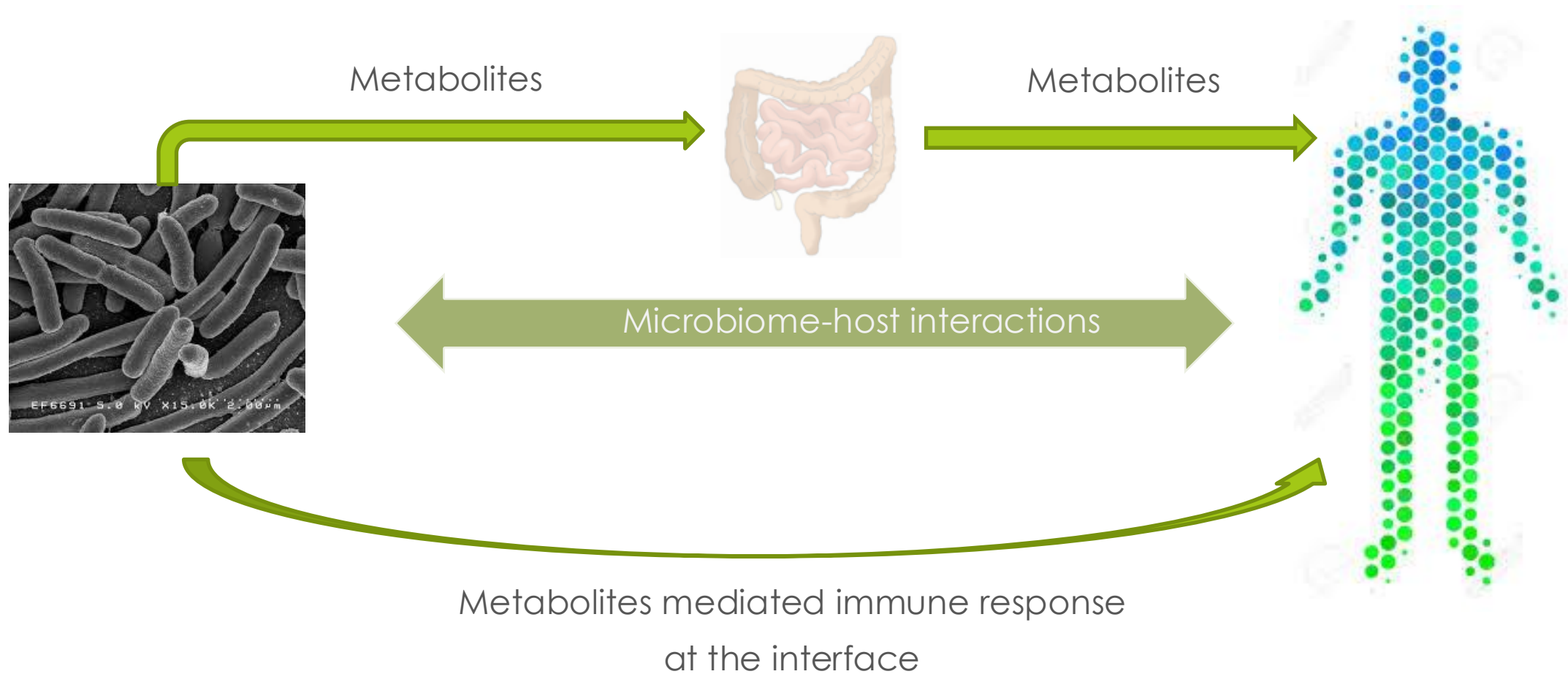
bacterial contamination



**Hematocrite level and
Collecte volume**



Stools: a window through the microbiota



Exploration of the metabolome



Reference value
Dynamic of the metabolome
Indivilualisation of the metabolome

Epidemiology and large scale studies

Take home message



Benchtop

Pre-Analytics
Sample choice
Sample quality
Collection conditions

Analytics
Targeted-
untargeted
Quantification

Post-Analytics
Reference values
population's
heterogeneity
Statistical models



- Robustness and reproducibility
- Quantification
- Easy validation and compatible with Quality system
- Well adapted to the study of large cohort



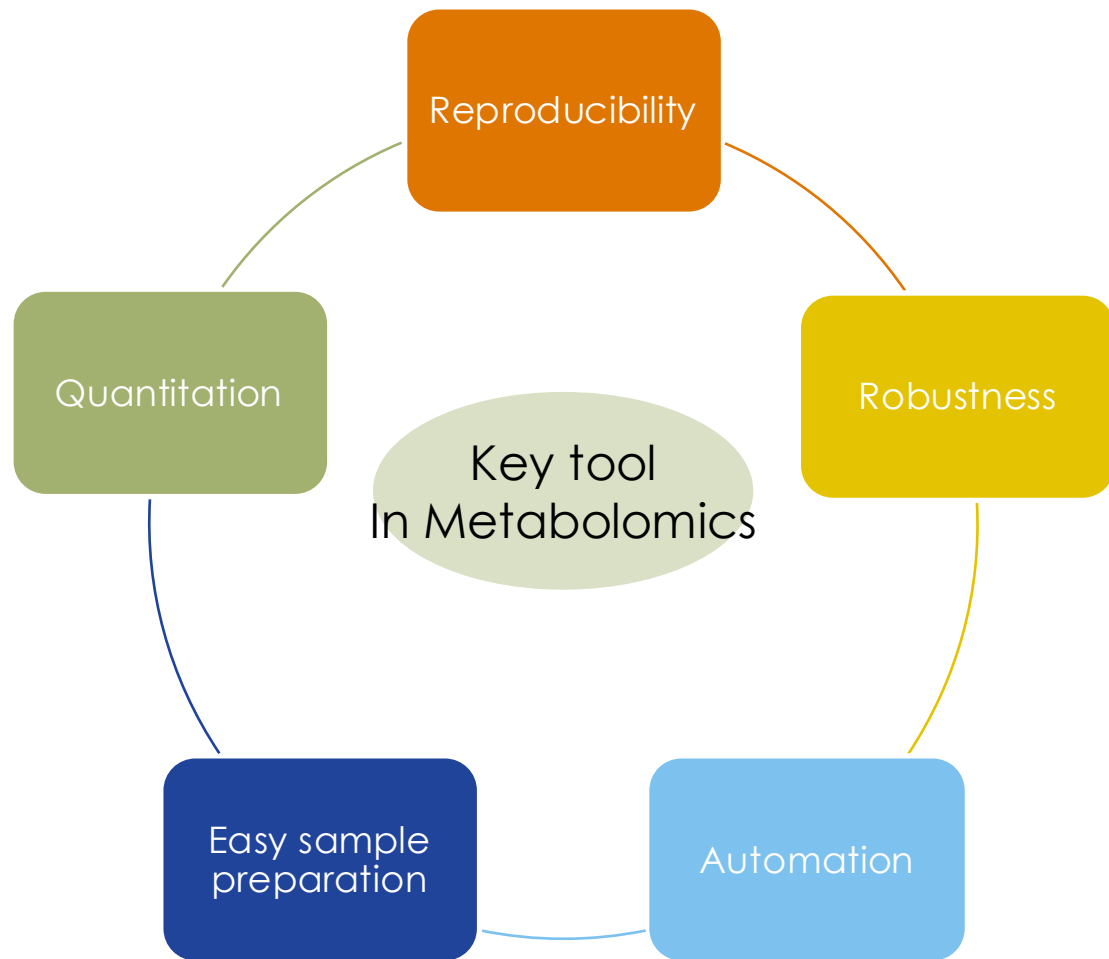
Clinics

Pre-analytics
optimisation

Exploration of new
matrices

Metabolome

Take home messages



Biomarkers discovery and validation

Epidemiology and huge cohorts

^{31}P studies

Fluxomics

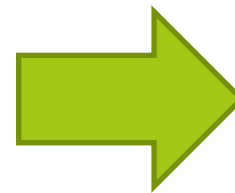
Lipoproteins profiling

Take home messages

Perfectly complementary with the SM

DNP and 2D spectroscopy are real opportunities to increase the potency of NMR in metabolomics but need to be developed

Choose the right analytical tool for the right biological question



Take home message

